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Paleomagnetic data from the Caborca terrane, Mexico: Implications for Cordilleran tectonics and the Mojave-Sonora megashear hypothesis

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Abstract. Two ancient magnetizations have been isolated in rocks of the Caborca terrane, northwest Mexico. The characteristic magnetizations of Neoproterozoic and Paleozoic miogeoclinal shelf-strata, arc-derived Lower Jurassic marine strata, and Jurassic volcanic and volcaniclastic rocks are of dual polarity and east-northeast declination (or south-southwest) and shallow inclination. Magnetizations in Neoproterozoic and Paleozoic miogeoclinal strata are interpreted as secondary (J*) and to be of similar age to those observed in Lower and Middle Jurassic rocks. Remanence acquisition is bracketed between about 190 and 160 Ma. The overall mean (D=15.0°, I=8.5°; n=38 sites; six localities; k=19.1, $\alpha 95$=5.5°) suggests a moderate to large clockwise rotation of 12 to 50° (depending on reference direction assumed) of the Caborca terrane, and rocks of the Sonoran segment of the Cordilleran volcanic arc, with respect to the North America craton. When compared with expected inclinations, observed values are not anomalously steep, arguing against statistically significant southward latitudinal displacement of the Caborca block after remanence acquisition. Late Cretaceous intrusions yield primary, dual-polarity steep inclination "K" magnetizations (D=341.4°, I=52.3°; n=10 sites; five localities; k=38.3, $\alpha 95$=7.9°) and have locally remagnetized Neoproterozoic and Jurassic strata. When present, secondary (K*) magnetizations in Neoproterozoic strata are of higher coercivity and higher unblocking temperature than the characteristic (J*) magnetization. Importantly, the regional internal consistency of data for Late Cretaceous intrusions suggests that effects of Tertiary tilt or rotation about a vertical axis over the broad region sampled (~5000 km^2) are not substantial. Late Cretaceous primary (K) magnetizations and secondary (K*) magnetizations yield a combined mean of D=348.1°, I=50.7° (N=10 localities; 47 sites; k=53.5, $\alpha 95$=6.7°), indicating at most small (<~10°) clockwise rotation of the Caborca region with respect to the craton. Permissible post-Late Cretaceous latitudinal displacement is near or below the detection limit of paleomagnetism (<~300 km). Limited data from Lower Cretaceous strata of the Bisbee Group (D=339.9°, I=47.9°; n=4 sites) suggest that the modest clockwise rotations inferred on the basis of J* magnetizations in Jurassic and older strata occurred in Jurassic time. Together, the lack of evidence for southward displacement, yet evidence for statistically significant clockwise rotation, and the overall similarity of Jurassic magnetizations in the Cordilleran arc with those of the Caborca block, despite the fact that some of them are clearly secondary, are not consistent with the Mojave-Sonora megashear hypothesis of Late Jurassic left-lateral strike-slip motion of the crust of northern Mexico.

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1. Introduction

The Caborca terrane of northwest Mexico, where Neoproterozoic and Paleozoic miogeoclinal strata are widespread (Figure 1a), has been interpreted as a para-autochthonous terrane displaced along the truncated western edge of the North America craton. *Anderson and Silver* [1979] proposed that the truncating structure is a Middle to Late Jurassic left-lateral strike-slip fault that they referred to as the Mojave-Sonora megashear (MSM). Their hypothesis builds on paleogeographic models for western equatorial Pangea that reconstruct Mexico north and west of its present position [e.g., *Van der Voo et al.*, 1974]. The MSM has become one of the most influential, yet controversial ideas regarding the Mesozoic evolution of Mexico and the Gulf region.

Evidence to support the MSM hypothesis includes the apparent juxtaposition of distinct Precambrian basement terranes in northwest Sonora [*Silver and Anderson*, 1974] and the similarity of the Neoproterozoic to lower Paleozoic miogeoclinal strata in Sonora with those of southeastern California and southwestern Nevada [*Stewart et al.*, 1984; *Ketner*, 1986]. Both arguments appear indisputable but do not provide conclusive evidence of mid-Mesozoic displacement [*Stewart*, 1988].

Provided statistically robust paleomagnetic data are collected, the hypothesis of southeastward displacement of the Caborca terrane can be independently tested with paleomagnetic techniques. The orientation, curvature, and sense of displacement of the proposed MSM predict that the Caborca terrane should have experienced modest counterclockwise rotation (15°) and small southward latitudinal displacement (~800 km) relative to stable North America. Paleomagnetic data for the Triassic-Jurassic Antimonio Formation, sampled west of Caborca and reported by *Cohen et al.* [1986], had been originally interpreted to support this model. A more detailed study of these rocks, however, conclusively reveals that the Antimonio Formation has been remagnetized in post-Sinemurian time. Magnetizations in rocks of the Antimonio Formation indicate that prior to the Late Cretaceous this area experienced a moderate clockwise rotation and this deformation is inconsistent with the megashear model [*Molina Garza and Geissman*, 1996a].

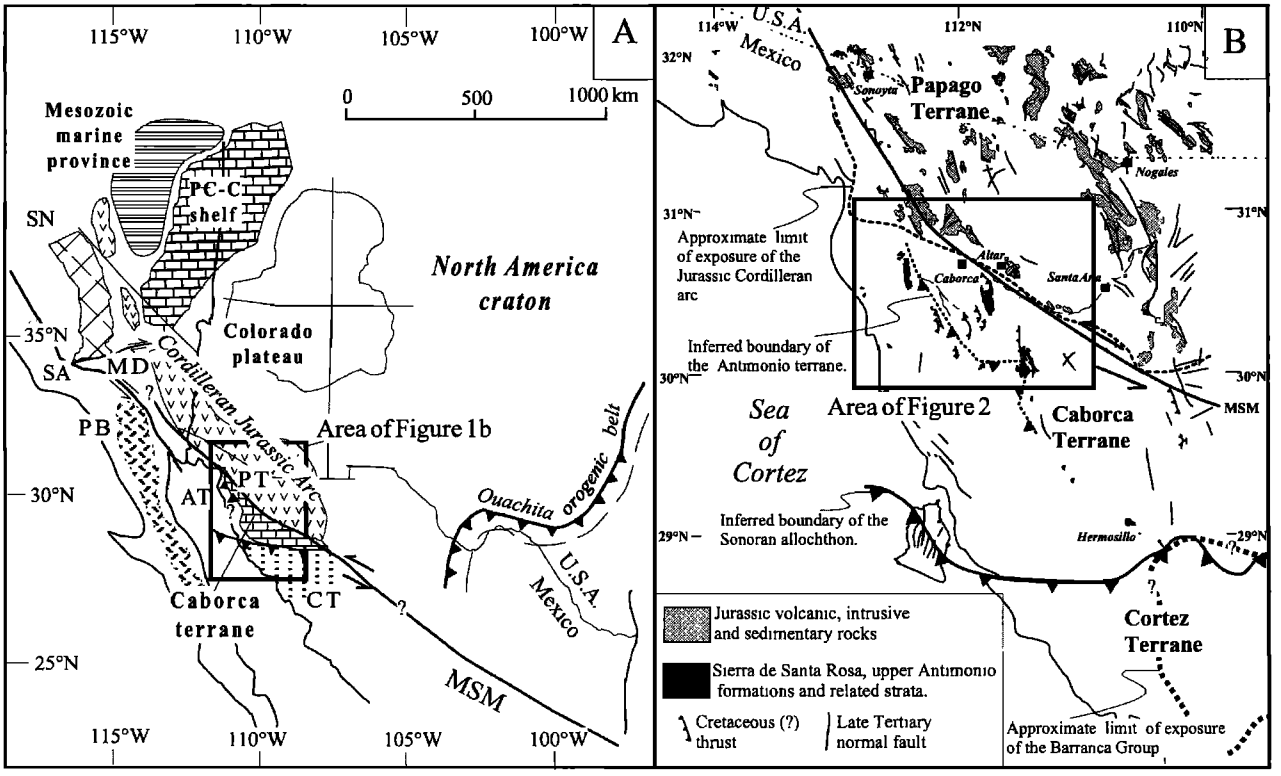


Figure 1. (a) Schematic tectonic setting of the Caborca terrane along the "truncated" southwest margin of North America south of the Cordilleran Jurassic arc. The Mojave-Sonora megashear model (MSM) links southeastward displacement of the Caborca terrane and northern Mexico with respect to the North American craton (identified by the outline of the Colorado plateau and the four corners region) to Jurassic continental truncation. In the MSM model, the Caborca and Antimonio terranes are interpreted as displaced fragments of the Precambrian-Paleozoic mioclinal Cordilleran shelf and the Mesozoic marine province, respectively, of eastern California and southern Nevada. The Caborca terrane is bounded to the south by the Cortez terrane (CT), a Paleozoic allochthon emplaced in the late Paleozoic or early Mesozoic. The relation between this allochthon and the late Paleozoic Ouachita suture is obscured by Tertiary volcanism. Late Cretaceous plutonism in the Caborca terrane is associated to emplacement of the Peninsular batholith, which in turn, has been truncated by the San Andreas fault in Neogene time. Symbols are as follows: AT, Antimonio terrane; PB, Peninsular batholith; SN, Sierra Nevada; SA, San Andreas fault; CT, Cortez terrane; PT, Papago terrane. (b) Distribution of lower Mesozoic rocks in western Sonora and southwest Arizona. Lower Mesozoic rocks belong to three distinct assemblages: (1) volcanic, intrusive, and sedimentary rocks of the Cordilleran arc (stippled), north of the inferred trace of the Mojave-Sonora megashear (MSM); (2) continental and transitional marine strata of the Barranca Group in the south and central areas of the state (the Barranca Group is exposed in the area bounded by a bold dotted line overlapping the Cortez and Caborca terranes); and (3) arc-derived marine and transitional marine strata of the Antimonio and Sierra de Santa Rosa Formations in isolated ranges between Caborca and Hermosillo (dark gray). These rocks have been assigned to the Antimonio terrane, but the areal extension of the Antimonio terrane remains unknown. The MSM is traced along the approximate limit of exposure of rocks of the Jurassic Cordilleran arc.

Here we report results of a more regional paleomagnetic study in the Caborca region. The structure and stratigraphy within the region are locally complex. This report emphasizes results from individual localities where structural relations among basement rocks, miogeocline strata, and Mesozoic rocks are relatively well understood and straightforward. Although some limitations exist in our present interpretation, simple tectonic models for the evolution of northern Mexico, like the MSM hypothesis of *Anderson and Silver* [1979] are difficult to reconcile with the paleomagnetic data.

2. Regional Geological Setting

2.1 Basement Rocks and Miogeoclinal Strata

The Caborca terrane [*Campa-Uranga and Coney*, 1983] is underlain by Paleoproterozoic basement of the Bamori Complex. Two U-Pb zircon dates of ca. 1.75 Ga [*Anderson et al.*, 1979] and younger (~1.65 Ga) U-Pb dates from pegmatites and K-Ar age determinations [*Damon et al.*, 1962] suggest that overall timing of crystallization, deformation, and metamorphism in the Bamori Complex is compatible with that of

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Precambrian provinces of southwest Arizona and eastern California. The Bamori Complex is overlain by Neoproterozoic through lower Paleozoic shallow water shelf-strata consisting of quartzite, dolomite, limestone, siltstone, and very minor amounts of conglomerate and volcanic rocks [*Stewart et al.*, 1984]. This miogeoclinal sequence is about 3 km thick and is unmetamorphosed but variably deformed, and carbonate rocks are often recrystallized.

The relations between the miogeoclinal sequence and relatively thin cratonal-platform strata in the far northeast and northwest regions of Sonora are not well understood, in part because regional facies and thickness changes have not been sufficiently studied. In contrast, to the south, in central Sonora, deep water Ordovician through Permian eugeoclinal strata of the Cortez terrane crop out along an east-west trending belt that constitutes the Sonoran allochthon (Figure 1). The allochthon is thrust on miogeoclinal strata [*Stewart et al.*, 1990; *Poole et al.*, 1995].

2.2 Mesozoic rocks

Precambrian basement rocks and miogeocline strata typically are in fault contact with variably deformed lower Mesozoic rocks that belong to three distinct assemblages: (1) intrusive, volcanic, and volcaniclastic rocks of the Cordilleran magmatic arc in the northern part of Sonora, north of the presumed trace of the inferred MSM [*Tosdal et al.*, 1990]; (2) continental and transitional marine strata of the Barranca Group in the south and central areas of Sonora [*Stewart et al.*, 1990]; and, (3) arc-derived marine and transitional marine strata of the Antimonio and Sierra de Santa Rosa Formations in western and northwestern Sonora [*Stanley and Gonzalez-Leon*, 1995]. Rocks of the Barranca Group and the Sierra de Santa Rosa and Antimonio formations lie south of the presumed trace of the inferred MSM (Figure 1b).

Variably deformed, sometimes metamorphosed, Lower through Upper Jurassic intermediate to silicic volcanic and Early to Late Jurassic intrusive rocks in northern Sonora represent the continuation of the Mesozoic continental arc of southeast California and southern Arizona and have been included in the Papago terrane [*Haxel et al.*, 1980; Calmus and Sosson, 1995; *Tosdal et al.*, 1990]. These rocks are well exposed in ranges along a northwest trending belt in the Altar desert region between Sonoyta and Santa Ana (Figure 1b), where they are affected by intense Late Cretaceous deformation. *Anderson and Silver* [1979] proposed that this outcrop belt of Jurassic volcanic rocks lies along the trace of the MSM. Volcanic rocks characteristic of the Papago terrane extend south, however, across the inferred trace of the MSM, to near the Sea of Cortez, where they are in the lower plate below thrusted Proterozoic gneisses [*Calmus and Sosson*, 1995]. North of Caborca at Sierra La Gloria (Figure 2), the volcanic-volcaniclastic sequence includes marine strata that contain Liassic ammonites (J. Garcia-Barragan, personal communication 1994). This indicates that a marine basin existed south of the volcanic arc.

Autochthonous sequences of Triassic and Lower Jurassic (?) strata in east and south-central Sonora include coal-bearing siliciclastic rocks of the Barranca Group [*Alencaster*, 1961]. The Barranca Group constitutes an overlap assemblage that postdates the emplacement of the Sonoran allochthon [*Poole et al.*, 1995]. The youngest strata of the Barranca Group, the Coyotes Formation, consists of a thick deposit of quartzite-boulder conglomerate that contains locally derived clasts of both deep and shallow water Paleozoic limestones.

Permian through Jurassic rocks of western Sonora are included in the Antimonio terrane (Figure 1b) and have been interpreted as an allochthonous sequence thrust on miogeoclinal rocks [*Stanley and Gonzalez-Leon*, 1995]. Petrologic and stratigraphic evidence suggest that the Antimonio terrane was located outboard of the continental margin but was probably associated with the early Mesozoic Cordilleran continental arc. Permian and Lower Triassic rocks of the Antimonio terrane have no correlative strata in the Caborca terrane and may indeed be allochthonous, but arc-derived strata of the Triassic and Lower Jurassic upper Antimonio Formation can be correlated with the Sierra de Santa Rosa Formation, and unnamed Mesozoic strata in Sierra Santa Teresa, near Hermosillo (Figure 1b). The conspicuous occurrence of a thick quartzite boulder conglomerate overlying marine and transitional Upper Triassic strata across western Sonora suggests that the Sierra Santa Rosa Formation and equivalent strata of the upper Antimonio Formation may correlate with the Coyotes Formation of the autochthonous Barranca Group [*Alencaster*, 1961; *Stewart and Amaya-Martínez*, 1993; *Lucas*, 1996].

Uppermost Jurassic (?) through Lower Cretaceous sedimentary rocks in northern Sonora consist of conglomerate, fine-grained siliciclastic rocks, limestone, and evaporites assigned to the Bisbee Group. Upper Cretaceous strata include siliciclastic rocks of the El Chanate Formation, and intermediate volcanic rocks of the El Charro Formation [*Jacques-Ayala et al.*, 1990; *Jacques-Ayala*, 1995].

East-directed contraction affects Proterozoic through Cretaceous strata of the Caborca terrane [*Rangin*, 1978; *De Jong et al.*, 1988]. The most intense deformation is concentrated along the inferred trace of the MSM [*Jacques-Ayala et al.*, 1990]. Late Cretaceous and early Tertiary (90 to 50 Ma) calcalkaline plutonic rocks of varied composition are widely exposed in Sonora and form the Sonoran (Laramide-age) Batholith [*Damon et al.*, 1983]. Cenozoic extension has modified older structures and imprints a characteristic basin and range physiography to areas of western Sonora. Mid-Tertiary core complexes developed in areas of north-central and east-central Sonora, primarily in rocks of the Mesozoic Cordilleran arc and the Bisbee Group [*Nourse et al.*, 1994]. Areas of large-magnitude extension were not sampled in the present study.

3. Sampling and Paleomagnetic Methods

In this study we sampled carbonate rocks of the Neoproterozoic and lower Paleozoic miogeocline sequence exposed in mountain ranges ("cerros" and "sierras") near Caborca (Figure 2). We also sampled Lower Jurassic volcanic, sedimentary, and volcaniclastic rocks; Lower Cretaceous sedimentary rocks; and intrusive rocks of Late Cretaceous age (Table 1). A sampling locality consists of four or more paleomagnetic sites collected from stratigraphic intervals tens to hundreds of meters thick. A typical site is an individual layer or a thin stratigraphic interval. Samples were drilled in the field and oriented using solar and magnetic compasses. Two

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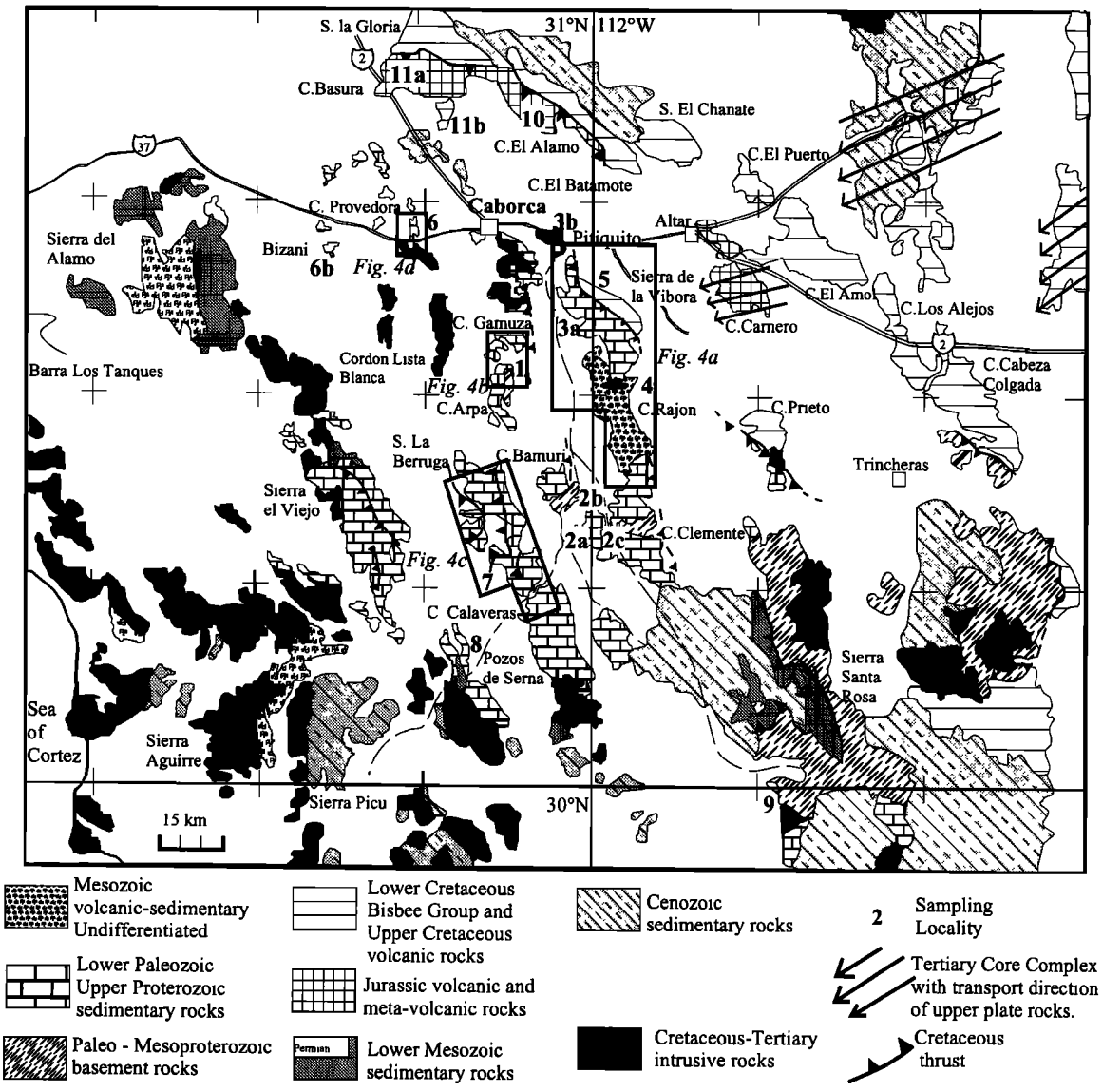


Figure 2. Simplified geologic map of the Caborca region [after *Merriam and Eells*, 1979; *De Jong et al.*, 1988; *Nourse*, 1995; and *Calmus and Sosson*, 1995], showing sampling localities (1-11), not specific sites. The locality numbers correspond to those listed in Table 1 and in the text.

or more specimens were prepared in the laboratory from each sample. Our conclusions are based on study of a total of 116 paleomagnetic sites collected over an area of about 5000 km^2. Sampling details are summarized in Table 1.

The materials collected show various degrees of recrystallization and alteration. Except for sites near Puerto el Alamo and Cerro Rajon (Figure 2), the rocks collected are not affected by penetrative deformation. In most ranges, strata dip at moderate to steep angles in well-exposed monoclinal sections. Tilt corrections were applied assuming a horizontal tilt axis (=strike). Map relations in the area appear to suggest that fold axes are horizontal. Except for those areas where Tertiary volcanic or sedimentary rocks are exposed, the effects of range-scale tilt associated with Basin and Range extension are difficult to evaluate with means other than paleomagnetic. As we will show below, the regional internal consistency of paleomagnetic results for Late Cretaceous intrusions and Miocene volcanic rocks suggests that relative rotation among localities is not important and structural tilt after remanence acquisition is only locally of concern.

Pilot samples were subjected to detailed alternating field (AF) or thermal demagnetization up to 120 mT or 695°C, respectively. A small number of samples in the collection contain low-coercivity magnetizations with either spurious isothermal magnetizations (IRMs) inconsistent within a site, possibly induced by lightning or during field sampling, or directions aligned with the present-day field (I=+49°). A smaller number of samples yielded uninterpretable results. In total, 10 sites were excluded for the above reasons, and the geologic significance of overall directions from 12 additional sites is unclear.

The vectorial composition of the natural remanent magneti-

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Table 1. Paleomagnetic Sampling in the Caborca Area

Locality*	Sites Collected*	Site Coordinates #	Rock Unit(s)	General Structure	Reference	Comments
Cerro Gamuza, Cerro el Arpa (1)	cg15-cg21	30.554°N-112.126°W	Caborca, Clemente,and Gamuza Fm.	NW dip (40° to 50°)	Damon et al. [1962]	Depositional contact between Precambrian strata and basement rocks
	ab22-ab24	30.531°N-112.116°W	Late Cretaceous intrusion			
	cab21-cab23 cg26-cg27	30.522°N-112.118°W	El Arpa Fm.	NW dip (40° to 50°)		
	ab24-ab25	30.517°N-112.117°W	1.1 Ga Aibo granite and mafic dike			
	cg28-30	30.512°N-112.141°W				
Lake Bamori, Cerro Clemente (2a, 2b, 2c)	bm40-41 (2a)	30.359°N-112.032°W	El Arpa Fm.	steeply to overturned WNW dipping strata	Anderson et al. [1979]	Precambrian strata in fault contact with basement rocks
	cc44-cc50 (2b)	30.300°N-112.009°W	Caborca and Clemente Fm.			
	cc39 (2c)	30.302°N-111.985°W	Caborca Fm.	SE dip~45°		
Sierra la Vibora (3)	sv51-53 (3a)	30.602°N-112.023°W	Papalote Fm.	NE dip, upper plate La Vibora	DeJong et al. [1988]	Intrusion post-dates thrusting of Precambrian on Mesozoic strata; depositional contact between Jurassic and mioclinal strata
	pit41-pit43 (3b)	30.695°N-112.063°W	Late Cretaceous Pitiquito intrusion.			
Cerro Rajon (4)	cr65-cr66	30.453°N-111.932°W	Late Cretaceous granodiorite		Longoria and Perez [1979]	
	cr67-cr70	30.453°N-111.926°W	Rajon Group (Jurassic?)	steep (~85°) SW dip	Stewart et al. [1984]	
Cerro la Ventana (5)	lv74-lv77	30.641°N-111.979°W	La Ventana Fm. (Cretaceous?)	steep NE dip		
	lv78-lv80	30.641°N-111.978°W	Tertiary volcanic rocks	gentle northeast dip		
Cerro Provedora (6)	cp54-cp59	30.700°N-112.255°W	Arrojos, Buelna, and Cerro Prieto Fm.	vertical-overturned, SSWstrike	Cooper et al. [1952]	
	cp81	30.680°N-111.270°W	Late Cretaceous intrusion			
Bizani (6b)	biz87-biz90	30 703°N-112.386°W	Devonian (?) and Mississip. Represo Formation and underlying strata	steeply (~60°) SW dip	Brunner [1975]	Isolated outcrops
Sierra la Berruga, Cerro Calavera (7)	cv31-cv37	30.270°N-112.126°W	Papalote and Cerro Prieto Fm.	NE-vergıng folds in upper plate of west dipping thrust	Eells [1972]	Folded Upper Precambrian strata
	cv38-cv39	30.280°N-112.100°W				
Pozos de Serna, Cerro Tilin (8)	til14-til19, til24	30.141°N-112.169°W	Lower Jurassic Sierra Santa Rosa Fm.	SW dip (60-70°)	Calmus et al. [1997]	Jurassic onglomerate lie on Precambrian strata and are thrust by Precambrian strata
	til26-30, 94	30.147°N-112.166°W	Lower Palozoic-Precambrian strata	vertical-overturned, SE strike		
	til20-23	30.152°N-112.185°W	Precambrian strata	upper plate rocks		
	til25	30.149°N-112.186°W	Late Cretaceous diorite intrusion			
	til91-til93	30.130°N-112.172°W	Tertiary volcanic rocks	apparent horizontal attitudes		
Sierra Santa Rosa (9)	sr31-sr32	30.009°N-111.723°W	Undifferentiated Precambrian	SE dip (~30°)		Thrust on Jurassic strata
Cerro Delicias	sr33-sr38	30.036°N-111.745°W	Lower Jurassic Sierra Santa Rosa Fm.	W dip (~45°)	Hardy [1981]	
Puerto Alamo (10)	cal42-cal53	30.850°N-112.010°W	Jurassic volcanic-volcanoclastic rocks	SW dip 50 to 90°	Nourse [1995]; Jacques-Ayala [1995]	North of the inferred trace of the MSM
Cerro la Basura (11a)	bas54-bas58	30.884°N-112.288°W	Lower Jurassic (?) Basura, marble unit	tightly folded	Corona [1979]; Nourse [1995]	
	bas70-73 (11b)	30.841°N-112.173°W	Red conglomeratic sandstone	NE di (~55°)		

* Numbers in parenthesis corresponds to locations in Figure 2.

Small discrepancies were found between GPS receiver coordinates and topographic maps for the area. For simplicity, when sites are scattered along a thick stratigraphic interval, the coordinates provided are for the first sampling station.

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zation (NRM) was determined from visual inspection of orthogonal demagnetization diagrams [*Zijderveld*, 1967]. The directions of magnetization components were calculated using principal component analysis [*Kirschvink*, 1980]. In many cases it was determined that a combination of demagnetization methods was more efficient in the separation of magnetization components. Site means were calculated using Fisher statistics. For less than 10 sites, mean directions were calculated using analysis of great circle trajectories, which were combined with linear data using the algorithm of *McFadden and McElhinny* [1988], without sector constraints. Paleomagnetic data and statistical parameters are summarized in Table 2.

4. Paleomagnetic results

4.1. Rock magnetic properties

To characterize the magnetic mineralogy of the rocks studied, IRM acquisition curves were obtained for representative specimens. Subsequently, specimens were subjected to AF demagnetization to examine the decay of the saturation IRM. Also, orthogonal IRMs were imparted with high (2 tesla (T)), intermediate (0.35 T), and low (0.12 T) inductions, and samples were subjected to thermal demagnetization, monitoring bulk-susceptibility changes during treatment. In addition, hysteresis curves were obtained for selected specimens. More details on the rock magnetic behavior of miogeoclinal and Jurassic strata will be published elsewhere.

IRM acquisition curves include a wide spectrum of responses, ranging from magnetite-dominated to higher-coercivity-dominated phases (Figure 3a). None of the miogeoclinal specimens studied completely saturate with inductions of 1.25 T. Specimens dominated by a single cubic phase (magnetite-type curves) have steep acquisition curves and nearly reach saturation with inductions of 0.25 T. An example typical of this type of behavior (cg19) is from a site where the only stable magnetization is of low coercivity and low unblocking temperature, in the direction of the present dipole field. Specimens dominated by high-coercivity phases also show concave upward IRM acquisition curves, which are typical of goethite. Some of the samples dominated by high-coercivity phases are red in color, others contain hematite bearing veins. Most samples are characterized, however, by mixtures of low- and high-coercivity magnetic minerals. Typically, IRM acquisition curves climb steeply below 0.1 T and display moderately to strongly pronounced inflections at around 0.25 T. Such a response can be produced by the precipitation of a high-coercivity phase (hematite?), by the oxidation of non magnetic (e.g., ferromagnesian) minerals, or by introduction of iron-bearing phases from distant sources. These curves cannot result from oxidation of magnetic phases of lower coercivity but higher saturation magnetization values. Introduction of new minerals is consistent with observations of hematite-bearing veins in different sedimentary rocks.

IRM acquisition curves suggest multiple generations of magnetic phases, which is consistent with the generally multivectorial composition of the NRM, as shown below. Intrusive and volcanic rocks are generally dominated by a cubic phase of low coercivity (magnetite?). Red sandstones are dominated by a high-coercivity phase (hematite?).

AF demagnetization of the NRM (Figure 3b) reveals that multidomain magnetite grains of low coercivity are characteristic of several rock types, as suggested by median destructive fields (MDF) of the NRM between about 10 and 15 mT (e.g., cg19, til23, and cc50). Behavior typical of intermediate coercivities is characteristic of other rocks, with MDFs of about 30 mT (e.g., cg15, cg30, bm41). This indicates that magnetite or maghemite carries a significant fraction of the natural remanence and low-coercivity magnetizations are likely to be viscous in origin. A large (>95%) to small (<1%) fraction of the NRM remains after demagnetization with AF inductions of 100 mT, indicating that hematite may also carry a significant fraction of the remanence.

4.2. Natural Remanent Magnetization

4.2.1. Overview. Progressive thermal and AF demagnetization of samples of Neoproterozoic through Lower Jurassic strata of the Caborca region isolates two ancient yet secondary magnetizations. One of them (the K* magnetization) is a magnetization of high coercivity and high laboratory unblocking temperature. K* magnetizations are of moderate to high inclination, and they are associated with Late Cretaceous magmatism. Late Cretaceous intrusions were sampled at five localities: Cerro el Arpa (labeled 1 in Figure 2), Cerro Pitiquito (3b in Figure 2), Cerro Rajon (4 in Figure 2), Pozos de Serna (8 in Figure 2), and Cerro Provedora (6 in Figure 2). A moderate- to high-coercivity magnetization (K magnetization) of moderate to high inclination was successfully isolated at all five localities, and directionally, it is indistinguishable from K* magnetizations in Neoproterozoic and Jurassic strata. Because they share a common mean, K and K* magnetizations are inferred to be of the same age and were combined in the final calculations. K* and K magnetizations are south-southeast directed and of moderate to steep negative inclination (or its antipode) and fail fold tests. K* magnetizations were observed at all of the localities studied (north and south of the trace of the MSM).

A shallow inclination (J*) magnetization was observed at four localities south of the trace of the MSM. Neoproterozoic strata at Cerro el Arpa (1 in Figure 2) and Cerro Clemente (2 in Figure 2) as well as Lower Jurassic strata at Sierra Santa Rosa (9 in Figure 2) and Pozos de Serna (8 in Figure 2) contain both K* and J* magnetizations, often in the same specimen. At Sierra de Santa Rosa and Pozos de Serna, J* magnetizations in Neoproterozoic strata are directionally indistinguishable from those observed in Lower Jurassic strata. Because they share a common mean, J* magnetizations in Neoproterozoic and Jurassic strata are inferred to be of the same age. J* magnetizations are south-southwest directed and of shallow inclination (or its antipode). They also fail fold tests.

Rocks of the Jurassic Cordilleran arc were sampled north of the trace of the MSM at Cerro la Basura (11a in Figure 2) and Puerto el Alamo (10 in Figure 2). These rocks have high-unblocking-temperature (>600°C) (J) magnetizations of dual polarity, which pass a paleomagnetic fold test. J magnetizations are interpreted to be primary in origin. J and J* magnetizations are interpreted to be of Jurassic age. As in the ranges south of Caborca, some of the sites collected in rocks of the Jurassic Cordilleran arc carry secondary magnetizations that are steeper than the characteristic magnetization and resemble

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Table 2. Paleomagnetic Data for the Caborca Region

	In Situ						Tilt Corrected	
Site	Dec, deg	Inc, deg	k	α95, deg	n\nd	Bedding	Dec, deg	Inc, deg
Cerros Gamuza and El Arpa								
Geomioclinal rocks								
cg15	204.6	-0.3	17.6	18.7	5\6	236,57	217.3	25.7
cg15 (K*)	161.4	-55.5			2\6			
cg16	213.3	3.7	12.8	26.7	4\6	236,57	226.4	21.0
cg17 (K*)	140.7	-58.1	108.4	15.8	3\7	236,57		
cg18	201.4	-2.0	99.1	12.4	3\5	234,54	211.7	24.5
ccg18 (K*)	177.7	-71.1			2\5			
cg20 (K*)	158.4	-54.7	54.9	10.4	5\5	220,41		
cg21	205.2	-4.0	297.3	4.4	5\5	220,41	206.1	6.6
cg21 (K*)	169.3	-53.8	74.3	8.9	5\5			
cg26 (K*)	188.4	-39.8	68.6	9.3	5\9	194,47		
cg26 o	49.8	66.2	18.1	18.5	5\9		316.8	52.8
cg27	24.1	-23.3	4.2	42.2	5\7	194,47	37.8	-8.7
cg27 o	35.2	50.9	24.0	16.0	5\7		339.0	44.1
cg27 (K*)	199.0	-48.6	26.5	24.4	3\7			
cg28 (K*)	188.8	-48.0	124.6	29.1	4\9 gc	212,40		
cg30	3.8	1.7	7.1	27.0	6\9	212,40	8.6	-16.3
cab21(K*)	205.0	-34.7	12.2	34.1	7\7gc	212,40		
cab22(K*)	185.8	-52.5	20.8	21.1	6\6 gc	212,40		
Cretaceous Diorite								
ab22	171.1	-67.8	160.4	6.1	5\5			
ab23	185.7	-49.7	18.2	16.1	6\6			
ab24	155.9	-46.6	129.7	6.7	5\5			
Precambrian dike								
ab25 o	38.3	61.9	31.6	10.9	7\9	212,40	332.0	48.2
Cerros Calavera and Aquituni								
Geomioclinal rocks west								
cv31 (K*)	187.4	-41.7	52.1	13.3	4\9	240,67	178.2	16.6
cv32 (K*)	18.8	46.6	163.5	6.3	6\7	240,67	1.4	-7.6
cv33 (K*)	3.2	53.8			2\7	180,50	314.6	33
cv34 (K*)	20.5	47.8	1005.3	2.4	5\7	191,36	343.4	41.6
cv35 (K*)	18.4	41.3	159.1	5.3	6\6	191,36	348.4	36.2
cv36 (K*)	13.2	47.3	104.2	7.5	5\10	191,36	339.8	37.6
cv36 o	142.7	-19.8	49.0	13.8	4\10		140.2	8.0
cv37 o	129.2	-25.0	64.6	8.4	6\10	282,38	138.0	-4.5
cv38 (K*)	324 7	67.5	35.3	13.0	5\8	240,16	326.7	51.5
cv38 o	113.7	15.4	9.4	20.8	7\11	240,16	109.7	28.0
cv39 (K*)	33.3	34.4	8.6	24.2	6\11	240,16	25.2	26.2

	In Situ						Tilt Corrected	
Site	Dec,deg	Inc,deg	k	α95,deg	n\nd	Bedding	Dec, deg	Inc, deg
Bamori-Cerro Clemente								
Geomioclinal rocks								
cc44 (K*)	179.2	-41.8	64.4	10.9	8\10	218,127		
cc45	13.8	20.4	16.2	17.1	6\10	210,127	22.3	-24.7
cc45 (K*)	170.9	-37.2	63.7	15.6	3\10			
cc46 (K*)	5.1	57.3	36.9	10.1	7\10	194,98		
cc47	12.3	22.6	10.8	29.2	4\8	206,105	6.7	-18.1
cc48	203.6	-22.0	12.6	36.2	3\8	206,105	185.2	7.7
cc49	6.3	21.5			2\7	206,105	8.7	-23.4
cc49 (K*)	192.8	-48.6	87.0	13.9	4\7gc			
cc50	186.9	-3.1	32.6	16.3	4\7	207/116.	213.4	19.3
cc39 (K*)	178.8	-46.0	93.8	7.6	8\8	181,52		
bm41	176.2	-3.8	11.0	25.8	5\8	338,60	170.6	13 7
bm40 o	109.8	44.1	92.7	9.6	4\7	338,60	96.8	-6.6
Sierra la Vibora and Cerro Rajon								
Cretaceous red beds Cerro la Ventana								
lv74	3.9	5.7	33.1	13.5	5\5	145,71.5	334.9	43.8
lv75	9.8	7.1	42.7	11.8	5\5	145,71.5	335.1	49.8
lv76	9.4	-0.1	66.5	9.5	5\5	145,71.5	345.2	47.1
lv77	12.3	1.4	266.9	5.6	4\4	145,71.5	344 8	50.4
Tertiary volcanic rocks Cerro la Ventana								
lv78	316.0	61.2	53.6	8.3	7\7	301,27	348.6	46.4
lv79	25.9	62.8	17.1	19.0	5\5	301,27	28.1	35.9
lv80	309.2	60.5	101.8	7.6	5\5	301,27	344.2	48.0
Cerro Rajon intrusive rocks								
cr65	328.7	48.1	71.7	7.2	7\10			
cr66	325.7	43.1	95.1	6.2	7\7			
Cerro Rajon Jurassic (?) sedimentary rocks								
cr67 (K*)	337.1	42.3			2\5	130,86	265.5	22.5
cr68 (K*)	329.1	39.7	18.8	15.8	6\6	112,78	251.3	35.9
cr69(K*)#	314.6	30.9	40.2	12.2	5\5	105,83	253.5	28.9
cr70(K*)#	340.6	34.8	316.0	4.3	5\5	105,83	238.8	47.9
Sierra Vibora Geomioclinal rocks								
sv51	17.2	20.2	10 7	29.5	4\7	320,32	19.2	-7.2
sv52 (K*)	164.1	-35.9	92 7	20.5	5\10gc	319,42	178.4	-18.4
sv53 (K*)	171.2	-47.2	89.1	23.6	5\11gc	319,42	189.4	-25.5
Cretaceous Pitiquito intrusion								
pit41	171.7	-51.5	26 7	11.1	7\16			
pit42	334.3	59.2	165.4	6.0	5\10			
pit43	152.4	-57.6	27.5	11.7	7\7gc			

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Table 2. (continued)

Site	In Situ Dec, deg	Inc, deg	k	α95, deg	n\nd	Bedding	Tilt Corrected Dec, deg	Inc, deg
Bizani geomioclinal rocks								
biz87(K*)	175.5	-54.5	58.9	21.8	6\10gc	89,76		
biz90(K*)	18.9	53.1	20.7	17.2	5\7gc	73,73		
biz90	274.5	73.5	254.2	7.7	3\7	73,73	179.6	22.3
Cerro Provedora Cretaceous								
cp81-int	323.6	50.0	50.8	10.8	5\7			
Cerro Provedora Geomioclinal								
cp54(K*)	301.8	52.3	29.9	12.5	6\7	29.5,92	117.8	35.7
cp55(K*)	314.6	53.2	129	8.1	4\6	29.5,82	108.7	33.4
cp56(K*)#	307.1	35 6	228.0	4 0	7\6	14, 106.5	81 5	33.5
cp57(K*)#	281 3	46 1	177.6	5 0	6\7	12.5,114	103.4	19.9
cp58(K*)#	280.1	29.0	46.3	9.0	7\8	25, 108	133 1	43.7
cp59(K*)#	288.3	33.3	209 4	5.3	5\6	10, 124	92.5	22.2
Cerro Tilin								
Tertiary								
til91	260.2	36.0	4.5	66.4	3\6			
til92a	341.1	56.1	13.9	25.5	4\4			
til92b	293.8	16.9	13.8	21 4	5\7			
til93	6.5	51.7	10.3	25.0	5\6			
Sierra Santa Rosa Fm.								
til14 (K*)	165.4	-45.6	28 0	17.7	4\11			
til15-16	358.5	28.7	101.0	9.2	4\12	164,74	318.7	20.1
til15 (K*)	183.8	-44.9	57.7	12.2	4\12			
til17	4 4	3 4	6.7	38.3	4\8	164,74	346.4	20.5
til17 (K*)	181 9	-43 2	102.5	6.6	6\8			
til18	223.6	5.2	21.5	16.9	5\6	164,74	190.7	-50.7
til18 (K*)	176 9	-45.4	70.7	11.0	4\6			
Geomioclinal rocks								
til21	15.2	13.4	17 1	15.0	7\10	175,83	343.3	21.2
til22a o	336 7	-75.3	136.3	7.9	4\8	174,92	70.0	-2.4
til22b	2.3	3.9	71.0	11.0	4\8		349.8	8.1
til23	0.7	12.2	12.7	16.1	8\8	152,90	318.2	28
til94	187.9	-13.8	14.2	21.1	5\5	164,76	155.3	-26.1
til26	2.8	22.2	429.0	3.2	6\7	138,97	283.3	37.0
til26 (K*)	166 4	-52 9	1306.0	3 4	3\7			
til27	1.5	13.7	525 5	2.4	8\8		294.7	39.7
til27 (K*)	174.7	-49 8	171.8	4.6	7\8			
til28	10.3	17.7	206 4	3.9	8\8		284.0	45.3
til29 o	308.8	4 5	17.6	18.8	5\7		314.6	-9.7
til30	7.1	19.3	130.5	5.9	6\8		284.1	41.9
Cretaceous stock Diorite								
til25	185.4	-39.3	21.3	27.0	6\8			
Sierra Santa Rosa								
Geomioclinal rocks								
sr31	212.4	6.6	45.3	13.8	4\9	44,32	216.0	-0.5
sr31(K*)	0.3	40.4	48.3	17.9	3\9	180,25	340.5	36.1
sr32	197.9	7.5	62.9	9.7	5\10	44,31	203.0	-6.9
sr32 (K*)	19.0	40.2	31 8	22.2	3\10	180,25	356.2	43.6
Sierra Santa Rosa Fm.								
sr33	15.2	13.9	10.3	30.2	4\7	185,63	355.7	15.2
sr33 (K*)	354.8	46.6	112 8	6 3	6\7	180,25	332.0	39.2
sr35	27.7	8.5			2\7	185,63	6.4	24.0
sr35 (K*)	210.5	-46.3	76 5	8.8	5\7	180,25	181.2	-53.5
sr36	42.2	-8.7			2\8	187,78	23.1	31.7
sr36 (K*)	39.6	44.2	14.3	18.3	5\8	180,25	12.2	55.6
sr37 (K*)	192.4	-37 4	18.1	18.5	5\6	184,50	172.5	-38.5
sr38 (K*)	0.9	33.6	132.1	6.7	5\5	184,50	334.7	18.7
Puerto el Alamo Jurassic								
cal42	355.1	-47.2	14.2	25.3	4\7	157,62	26.2	-9.1
cal43	353.9	-50.5			2\2	157,62	29.0	-10.8
cal44	353.7	-30.0	29.6	12.5	6\8	165,67	15.9	-4.3
cal45	333.0	-28.4	16.7	19.3	5\6	144,69.5	353.6	-2.2
cal46 (K*)	352.7	49.1	18.8	13.1	8\8	162,34	315.4	44.0
cal47	230.5	72.0	51.3	9.4	6\6	130,66	223.2	6.3
cal48	33.8	-48.8	8 3	28.2	5\7	128,66	35.2	7.1
cal49 o	201.8	7.2	16.4	17	6\6	128,56	194.5	-46.0
cal51 o	358.1	6.1	8.6	18.6	9\11	142,89	322.5	36.5
cal52 (K*)	5.6	60.0	101.3	6.7	6\8	117,39	347.7	39.0
cal53 o	270.2	42.4	15.4	15 9	7\7	116,34	251.9	22.3
Cerro Basura intrusions								
bas54	342.8	23.2	97.9	6.6	6\6	320,35	351.4	6.8
bas55	349.4	24.7	36.8	16.0	7\7	320,35	357.4	5.0
bas56	183.6	-4.9	11.0	21.3	5\5gc	148,29	177.9	-20.8
bas57(K*)	136.9	-43.8	11.9	20.3	6\6			
bas58	6.5	31 1	71 8	6.6	8\13	17,23	21.0	32.4
Cerro basura red beds								
bas70(K*)	179.2	-56.8	1710.9	3.0	3\4	318,55	203.0	-10.3
bas71(K*)	173.7	-56.4	196.0	6.6	4\4	318,55	200.6	-12.3
bas72(K*)	180.4	-57.1	228 6	3 7	8\8	318,55	203.9	-10.5
bas73(K*)	172.8	-54.0	221.8	6.2	4\4	318,55	198.6	-10.9

Dec and Inc are declination and inclination of the site mean; k and α95 are the Fisher distribution parameters; site means are listed in situ and after tilt correction; n\nd is the ratio of samples used in the calculations/samples demagnetized Bedding is the attitude (strike and dip) of layering. Cretaceous overprints are labeled K*; "o" indicates magnetizations not readily interpretable. # Sites excluded from the locality mean and the final calculations.

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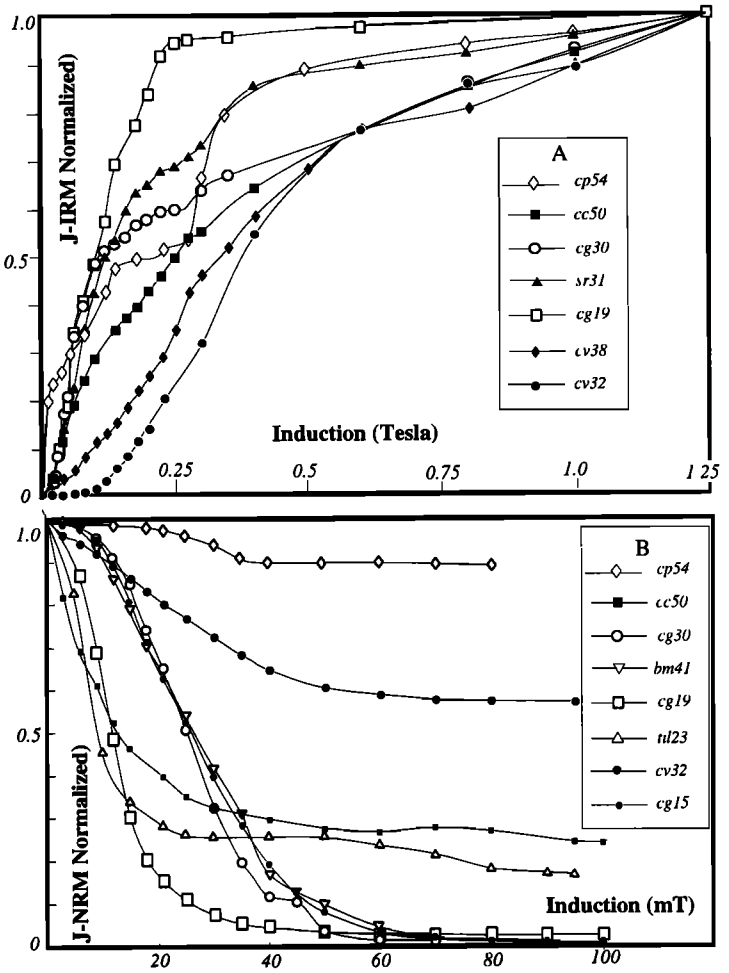


Figure 3. (a) Acquisition curves of isothermal remanent magnetization for selected specimens of the miogeoclinal section of the Caborca region. (b) Decay of the intensity of the natural remanent magnetization of selected specimens of Neoproterozoic miogeoclinal carbonate rocks during alternating field demagnetization.

directions observed in Cretaceous intrusions. They are thus interpreted as K* magnetizations. A detailed description of paleomagnetic data follows; results are summarized in Tables 2 and 3.

4.2.2. Localities south of the inferred trace of the Mojave-Sonora megashear. Details of the paleomagnetic results for sampling localities south of the inferred trace of the MSM (localities 1-9; Figure 2) are described below.

4.2.2.1. Cerro el Arpa-Cerro Gamuza: Cerro el Arpa (Figure 2, locality 1) is a small range where west-dipping Neoproterozoic strata overlie in depositional contact the 1.1 Ga Aibo granite [*Damon et al.*, 1962]. These rocks are intruded by a fine-grained Late Cretaceous stock of intermediate composition (Figure 4a). The Late Cretaceous microdiorite contains a reverse polarity (south directed and moderately steep negative) characteristic magnetization (ChRM) of moderate to high coercivity (> 130mT) overprinted by randomly-oriented low-coercivity IRMs (Figures 5a and 5b). Thermal and AF demagnetization isolate an identical ChRM, but thermal demagnetization is less effective in removing the overprinting magnetization. Laboratory unblocking temperatures of less than 590°C suggest that the remanence of the diorite is carried by magnetite. IRM acquisition curves of samples of this intrusion reach saturation with inductions of about 0.22 T. We refer to this magnetization as the K magnetization. This is interpreted as a primary thermoremanent magnetization (TRM).

Demagnetization behavior of miogeoclinal carbonate rocks at Cerro el Arpa and Cerro Gamuza is relatively straightforward. The ChRM is a shallow south-southwest directed (J*) magnetization isolated at temperatures between 240 and 550°C or inductions of 20 to 130 mT (Figure 5c). The ChRM is overprinted by a low-coercivity and low-unblocking-temperature magnetization (less than about 250°C) of direction similar to the recent dipole field (Figure 5d). A third magnetization (K*) of high coercivity and intermediate to high unblocking-temperature is south directed and of steep negative inclination (Figure 5d). This (K*) magnetization is essentially identical to the ChRM of the Late Cretaceous microdiorite (Figures 5f and 5g). The K* magnetizations become more prominent near the Cretaceous intrusion; in five sites (Table 2), this is the characteristic magnetization. The high unblocking temperature (>625°C) and coercivity (>100 mT) indicate that the steep negative south directed K* magnetization re-

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Table 3. Summary of Paleomagnetic Directions for the Caborca Region

Locality	K/K* Magnetizations						In Situ ChRM (J/J*)					Tilt Corrected ChRM			
	n (nK/nK*)	Dec, deg	Inc, deg	*k*	α95,deg	Overall	*n*	Dec, deg	Inc, deg	*k*	α95,deg	Dec, deg	Inc, deg	*k*	α95,deg
Cerro del Arpa	13 (3/10)	177.8	-53.8	28.8	7.9		6	202.1	3.1	33.1	11.8	211.3	17.5	30.1	12.4
Cerro Clemente	5 (0/5)	180.7	-46.1	73.2	9.0		6	9.6	15.7	39.7	10.8	10.9	-18.3	27.6	13.0
Sierra La Vibora	5 (3/2)	163.6	-50.6	55.7	10.3		1	20.2	17.2			19.2	-7.2		
Cerro Provedora	3 (1/2)	313.5	52.2	134.7	10.7										
Bizani	2 (0/2)	187.4	-54.4				1	274.5	73.5			179.6	22.3		
Cerro Calavera	8 (0/8)	13.6	48.6	29.2	10.4										
Cerro Rajon	4 (2/2)	330.2	43.4	262.2	5.7										
La Ventana red beds †							4	8.9	3.5	272	5.6	339.9	47.9	271.9	5.6
La Ventana Tertiary							3	334.2	65.8	18.4	29.6	1.8	45.3	18.4	29.6
Santa Rosa ‡	7 (0/7)	349.9	43.4	38.0	9.9		2+3	204.9	-11.0	28.8	14.5	197.0	-16.1	16.9	19.2
Cerro Tilin-Pozos de Serna	7 (1/6)	176.8	-46.1	129.9	5.3		8+3	8.5	13.3	27.7	8.8	322.1	34	8.6	16.5
Puerto el Alamo	2 (0/2)	358.3	54.7				6	359.5	-48.1	12.0	20.1	24.0	-4.4	19.5	15.6
Cerro la Basura	5 (0/5)	167.1	-54.9	41.7	12.0		4	355.6	21.3	28.0	17.7	1.2	16.5	21.3	20.4
Locality mean (62 sites)	*N*=11	350.8	51.0	48.2	6.6										
Normal	*n*=24	350.3	50.6	19.5	6.9	Mean 1	n=28	15.0	10	23.0	5.8	11.5	8.1	4.1	15.2
Reverse	*n*=38	175.8	-51.1	43.0	3.6	Mean 2	n=10	173.7	21.4	3.7	28.9	15.0	4.0	12.4	14.3
Overall site mean	*n*=62	353.7	50.3	29.5	3.4	Overall	n=38	15.0	8.5	19.1	5.5				
Intrusive rocks only	*n*=10	341.4	52.3	38.3	7.9	Selected	n=24	13.5	9	18.6	7.1				
Selected mean (47 sites)	*N*=10	348.1	50.7	53.5	6.7	Locality	N=6	15.2	8.3	35.8	11.3				

Here *n (N)* is the number of sites (localities) used in the locality mean calculation, (K/K*) indicates the ratio of primary (K) versus secondary (K*) magnetizations used. Dec and Inc are the mean declination and inclination, *k* and α95 are the statistical parameters of the Fisher distribution. In the final calculations, Mean1is the mean of J* magnetizations south of the inferred trace of the MSM and Mean 2 is the mean of J magnetizations north of it. The means are calculated either assigning unit weight to each site mean (the number of sites is listed as "*n*"),or assigning unit weight to each locality (the number of localities is listed as "*N*"). Also, in situ magnetizations south of the trace of the MSM are combined with tilt-corrected site means north of it. Selected means include sites of $n>2$ samples, $k>10$, and $\alpha95<20°$.

† Tilt corrected directions include a correction for the tilt of overlying Miocene volcanic rocks.

‡ All directions are corrected for the westward tilt of overlying Tertiary conglomerate.

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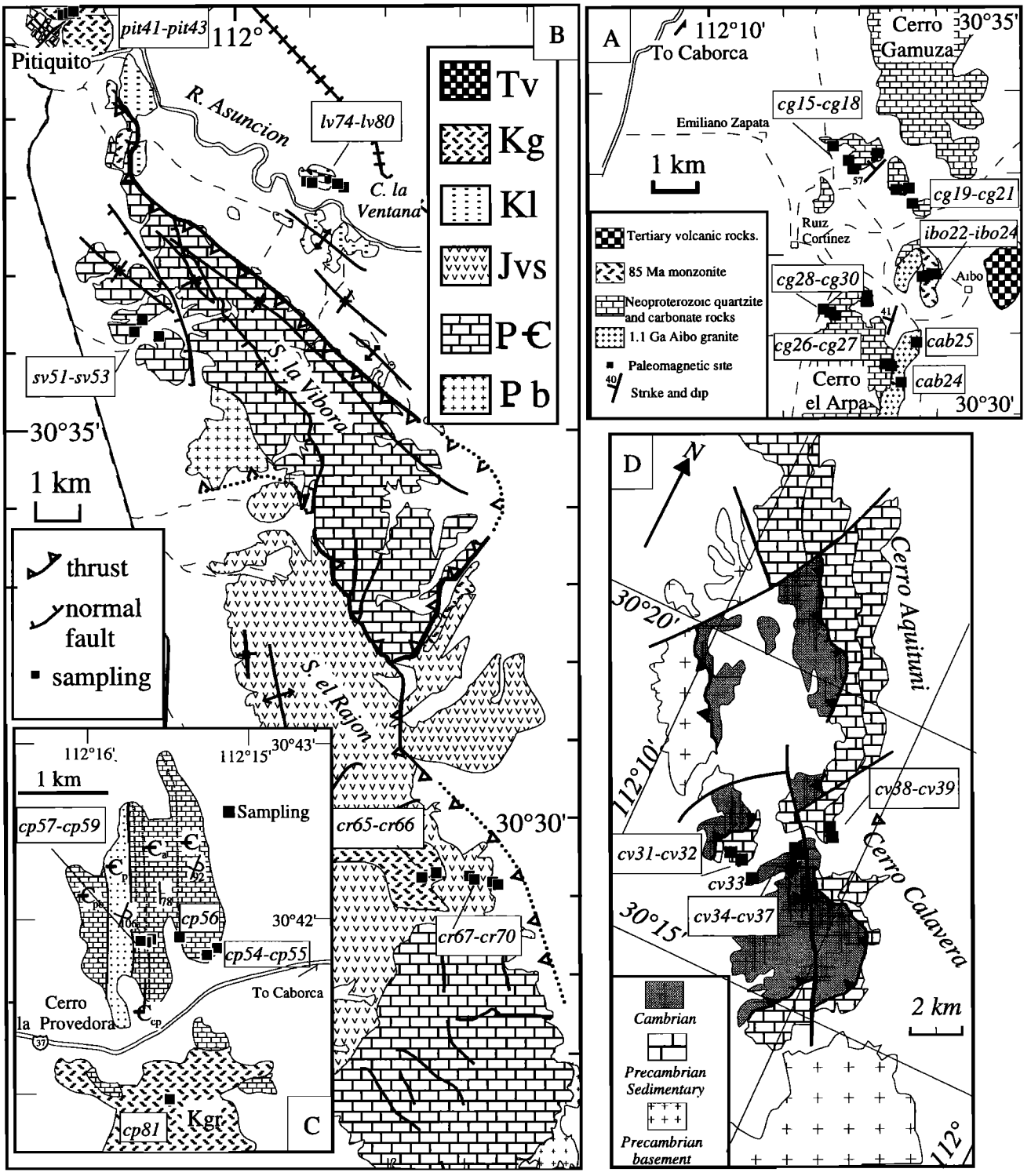


Figure 4. (a) Simplified geology of Cerro el Arpa and Cerro Gamuza [after *Damon et al.*, 1962], showing sampling sites, as squares, in this and other figures. (b) Simplified geology of Sierra la Vibora and Sierra Rajon [after *De Jong et al.*, 1988; *Longoria and Perez*, 1979; and *Stewart et al.*, 1984], showing sampling sites. (c) Simplified geology of Cerro Provedora [after *Cooper et al.*, 1952], showing sampling sites. (d) Simplified geology of Sierra la Berruga [after *Eells*, 1972], showing sampling sites.

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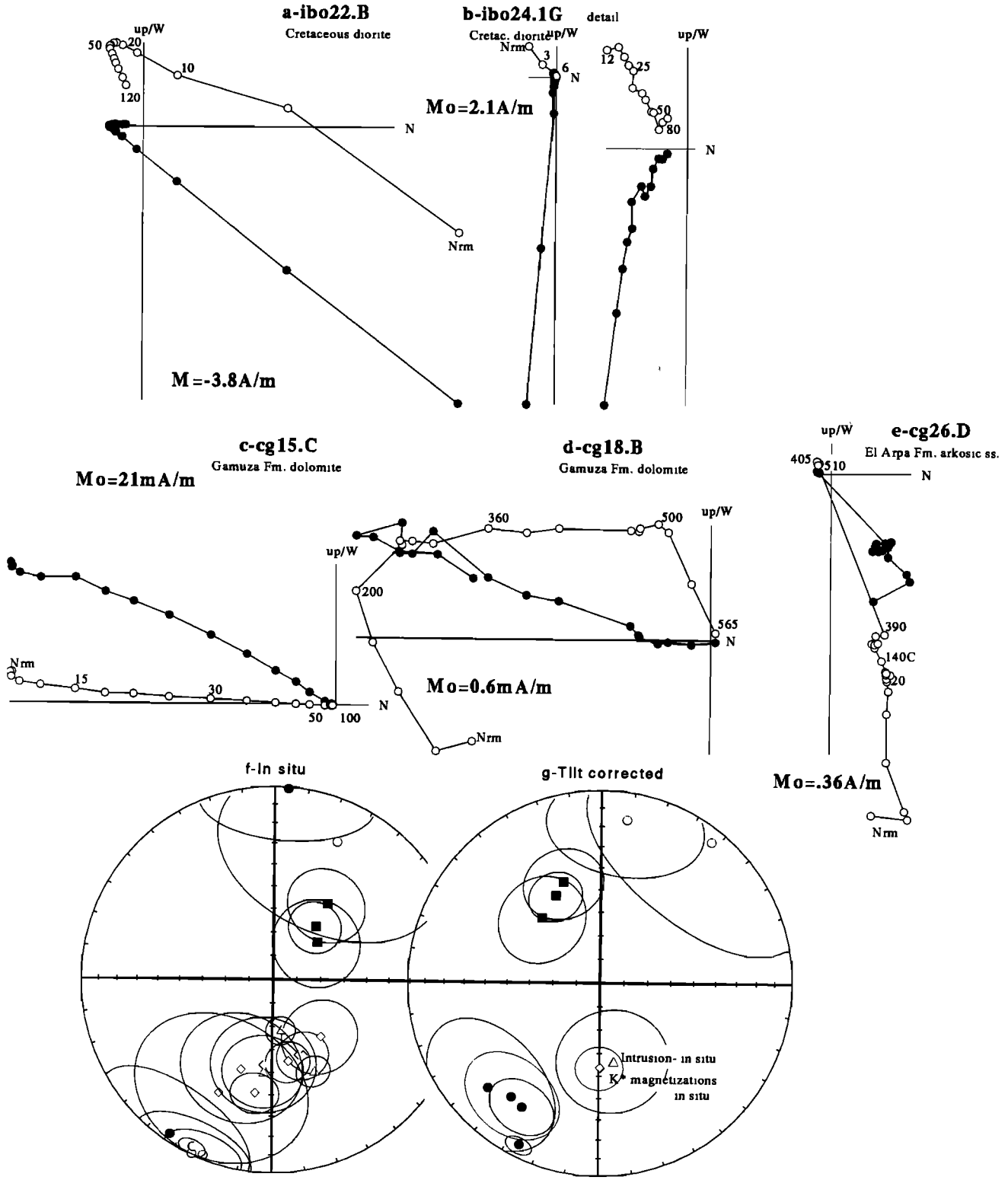


Figure 5. (a-e) Orthogonal demagnetization diagrams of samples collected at Cerro el Arpa and Cerro Gamuza. Open (closed) symbols are projections in the vertical (horizontal) plane. Temperatures are in degrees Celsius; Figures 5f and 5g are equal-area projections (in situ and tilt corrected, respectively) of site means and corresponding 95% cones of confidence for the characteristic J* magnetizations in stratified rocks (circles), a Cretaceous intrusion (triangles), K* remagnetized miogeoclinal rocks (diamonds), and other directions (squares). Closed (open) symbols are projections on the lower (upper) hemisphere. See the text for additional explanation.

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sides in hematite. IRM acquisition curves for this locality indicate the presence of magnetic phases of both low and high coercivity, interpreted as magnetite and hematite. The steep south directed K* magnetization in Neoproterozoic strata is interpreted to be a Late Cretaceous secondary magnetization of thermochemical or chemical origin (TCRM or CRM). The combined in situ mean of K and K* magnetizations at this locality is D=177.8°, I=-53.8° (n=13 sites; Table 3). The J* magnetization is shallow in both in situ and tilt-corrected coordinates (Figures 5f and 5g). In some cases, within-site scatter is large (Table 2), but between site dispersion is low, with an in situ locality mean of D=202.1°, I=3.1° (n=6 sites; Table 3).

Hematitic sandstones and carbonate rocks near the nonconformity between the Aibo granite and the Neoproterozoic sequence contain high-temperature K* overprints and a different magnetization of high-coercivity but intermediate unblocking temperature. This magnetization is northeast directed and moderately steep positive, and it is unblocked over a narrow range of laboratory temperatures between 370 and 400°C, thus removed before the south directed and moderately steep negative K* component (Figure 5e). An identical magnetization was isolated in an altered diabase dike (site ab25) that intrudes the Aibo granite but not the sedimentary sequence. In situ, this magnetization does not resemble the present or the Tertiary field directions (D=40.1°, I=59.8°; n=3 sites). In tilt-corrected coordinates the magnetization is northwest directed and moderately steep positive (D=329.9°, I=48.7°). The correct reference frame for the characteristic, shallow south-southwest directed J* magnetization, and the steep, northeast directed magnetizations cannot be uniquely determined at this locality. Three sites (cg19, cg29, and cab23) did not produce interpretable results. Also, a site in the Aibo granite yields univectorial magnetizations of moderate coercivity and high unblocking temperature, with no internal consistency. We regard these as spurious.

4.2.2.2. Cerro Clemente and Bamori Ranch: At Cerro Clemente (Figure 2, locality 2), outcrops of steeply dipping Neoproterozoic strata are dominated by siltstone, sandstone, and minor dolomite of the Clemente Formation, and by carbonate rock of the underlying Caborca Formation [*Stewart et al.*, 1984]. Structural relationships with basement exposed at Bamori Ranch, 2 km to the north, are masked by Quaternary alluvium. Seven sites were collected on the west side of Cerro Clemente (cc44-cc50) and one on the east side (cc39). Two additional sites were collected near Bamori lake (bm40 and bm41). Carbonate rocks at Bamori lake occur in (steep) fault contact with quartz-feldspathic gneisses of the Bamori Complex.

The ChRM of Proterozoic strata at Cerro Clemente is a dual polarity J* magnetization (south or north directed) of shallow inclination (Figures 6a and 6b). The J* magnetization is overprinted by a magnetization in the direction of the recent dipole field and is characterized by intermediate to high coercivities and unblocking temperatures of less than 500°C. Two sites yield only south directed (K*) magnetizations of moderate to steep negative inclination, with coercivities above 100 mT. At one site a K* magnetization of normal polarity was observed. Samples from sites cc45 and cc49 contain both the shallow northeast directed magnetization and the steep south directed and steep negative K* magnetization. In site cc45 the J* magnetization is of both polarities, a south-southeast directed magnetization is removed before the steep K* overprint is isolated (Figure 6c). Cretaceous intrusions do not crop out in the Cerro Clemente area.

IRM acquisition curves of samples from Cerro Clemente are dominated by high-coercivity phases (Figure 3), but the presence of a cubic phase is evident from the steep climb of the IRM curve below 0.1 T. Furthermore, about 80% of the saturation IRM is removed in inductions of 20 mT. The ChRM of sites such as cc50 (Figure 6b) apparently resides in a high-coercivity phase of low unblocking temperature, such as fine-grained hematite.

Samples from site bm41 display a north directed steep positive magnetization that unblocks below 350°C; upon further treatment, a west directed and moderately steep magnetization unblocks below 550°C (Figure 6d). AF response shows that this magnetization is soft and possibly spurious. The AF demagnetization trajectories trend toward a south directed end-point (Figures 6e and 6f). Combined end-points and great circle trajectories yield a well-defined shallow south directed site mean similar to the ChRM in sites from Cerro Clemente (Figure 6f; Table 2). Site bm40 yields (in situ) east directed magnetizations of moderate inclination (Table 2). The J* magnetization is shallow in both in situ and tilt-corrected coordinates (Figures 6f and 6g), but as in Cerro el Arpa, a fold test is inconclusive. Also, within-site scatter is large (Table 2, Figure 6f), but overall, the mean locality is well defined. In situ, the ChRMs yield a locality mean of D=9.6°, I=15.7° (n=6 sites; Table 3).

4.2.2.3. Sierra la Vibora: Sierra la Vibora and Sierra Rajon consist of exposures of Proterozoic crystalline basement, Neoproterozoic and Paleozoic miogeoclinal strata, Mesozoic volcaniclastic rocks, and Cretaceous intrusive rocks (Figure 4b). Miogeoclinal strata are thrust on folded Mesozoic strata [*De Jong et al.*, 1988]. The Late Cretaceous Pitiquito granodiorite and a diorite stock at Cerro Rajon were emplaced after folding and thrusting.

The Pitiquito intrusion contains a moderately steep ChRM of moderate to high coercivity (>130 mT) overprinted by spurious low-coercivity IRMs (Figures 7a and 7b). Both polarities are present (Table 2), and three sites give a mean K direction of D=340.2°, I=56.4°. Laboratory unblocking temperatures less than 590°C suggest that the remanence is carried by magnetite. The ChRM in site sv51 is similar to the north-northeast directed J* magnetization observed in Cerro Clemente and Cerro Gamuza (Figure 7c) but is defined by a single site and was excluded from the final calculations. Two sites in miogeoclinal strata (sv52 and sv53) yield southeast and moderately steep negative K* magnetizations overprinted by a prominent overprint in the direction of the recent dipole field (Figure 7d). Low magnetic moments make it impossible to resolve end-point vectors in all samples (Figure 7e), but combined demagnetization great circles and end-point vectors define site means with directions near the mean of the Pitiquito intrusion, although with shallower inclinations (Figure 7f). Although the small difference in inclination between sites sv52 and sv53 and sites collected in the intrusion is not statistically significant, it could be attributed to either incomplete removal of the steep north directed overprint in the miogeocli-

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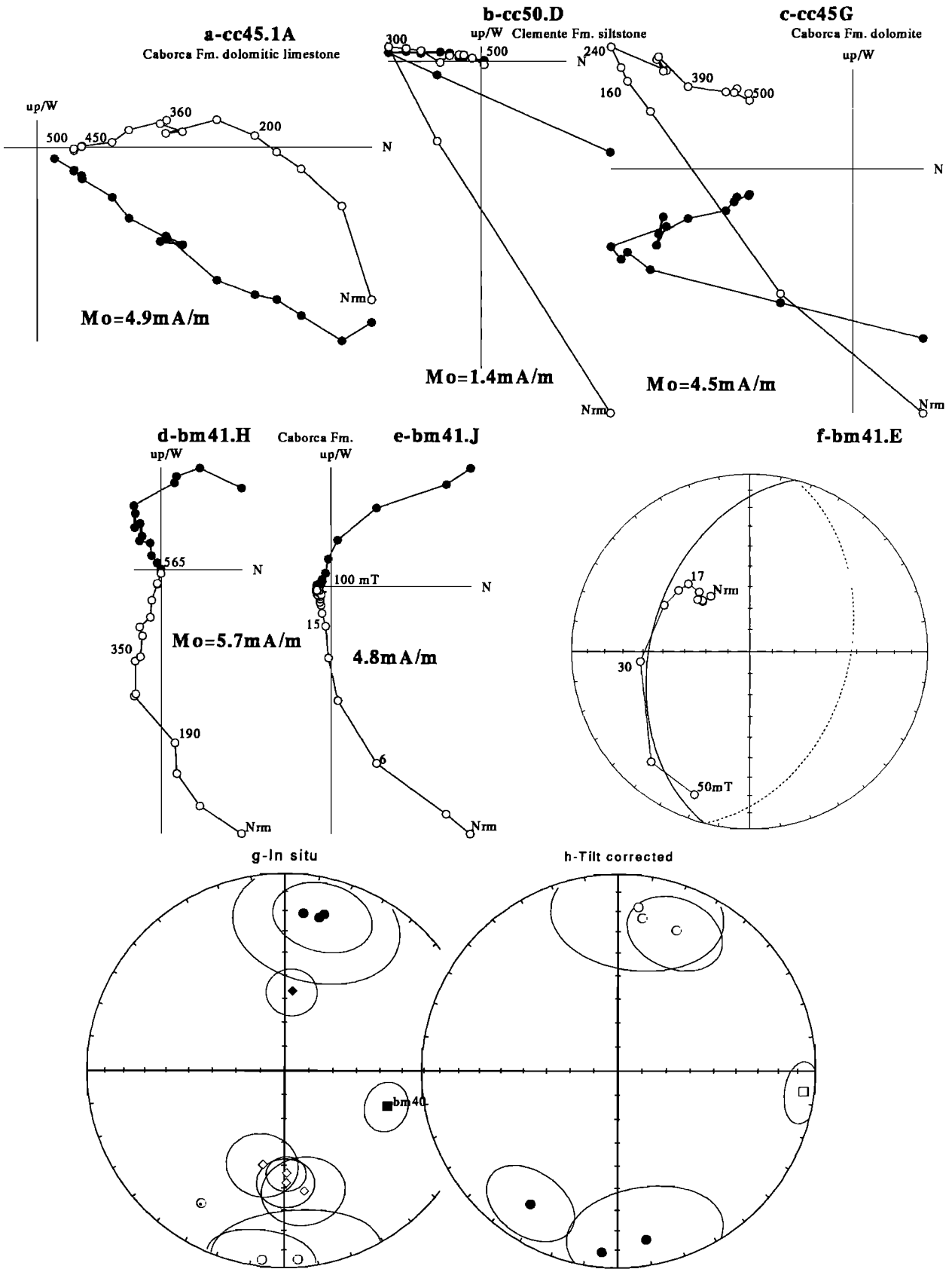


Figure 6. (a-e) Orthogonal demagnetization diagrams of samples collected at Cerro Clemente and ranch Bamori. Temperatures are in degrees Celsius and inductions in milliteslas. Figure 6f shows an example of a great circle demagnetization trajectory of a sample from site bm41. Figures 6g and 6h are equal-area projections of site means (in situ and tilt corrected, respectively) and corresponding 95% cones of confidence for the characteristic J* magnetization in stratified rocks (circles), K* remagnetized miogeoclinal rocks (diamonds), and other directions (squares). Other symbols are as in Figure 5.

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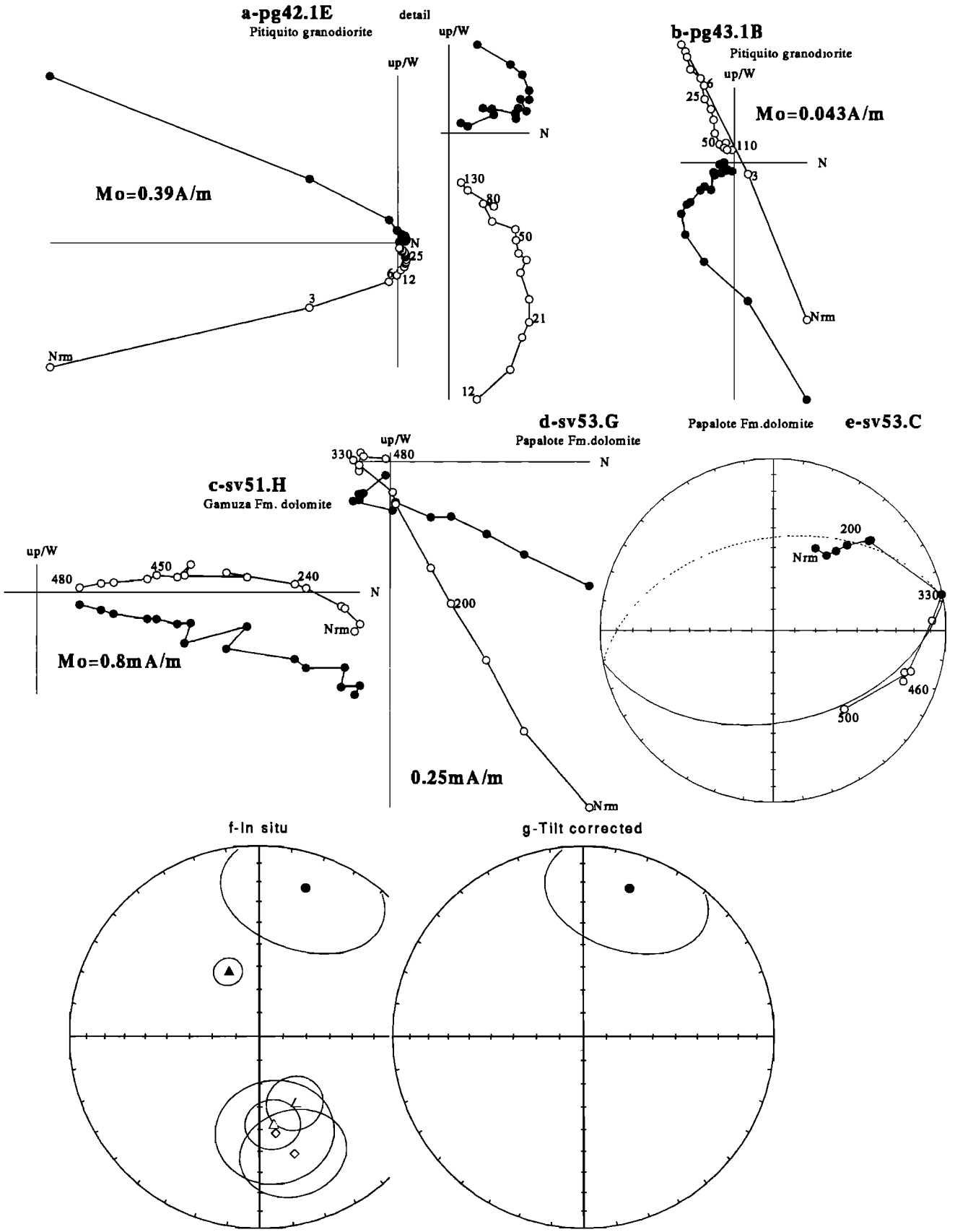


Figure 7. Orthogonal demagnetization diagrams of samples collected at (a, b) Pitiquito and (c, d) Sierra la Vibora. Figure 7e shows an example of a great circle demagnetization trajectory of a sample from site sv53. Symbols are as in Figure 5. Figures 7f and 7g are equal area projections (in situ and tilt corrected) of sites means and corresponding 95% cones of confidence for the characteristic magnetization (circles), K* remagnetized miogeoclinal rocks (diamonds), and the Late Cretaceous Pitiquito intrusion (triangles).

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nal rocks or to gentle east-side downtilt (after remanence acquisition) of the area where sites sv52 and sv53 were collected. Combined, the K and K* magnetizations have a mean of D=163.6°, I=-50.6° (Table 3).

4.2.2.4. Cerro Rajon: The Cerro Rajon stock intrudes Jurassic (?) light gray quartz arenite, volcaniclastic sandstone, and green-purple phyllite of the Chino Group (Figure 4b). The Rajon stock yields northwest directed and moderate positive inclination K magnetizations (Figure 8a) overprinted by low-coercivity magnetizations in the direction of the recent dipole field. Mesozoic strata adjacent to the intrusion contain similar magnetizations (Figure 8b), which are interpreted as secondary (K*) TRMs or TCRMs associated with the intrusion. Remagnetized Jurassic rocks yield, however, systematically shallower inclinations than the intrusion (Figure 8c). Two sites in phyllites (cr69 and cr70), farther from the intrusion, yield the shallowest inclinations. Demagnetization trajectories for these sites show that an unresolved, shallow magnetization is present in the high-temperature demagnetization steps (Figure 8b). It appears that the K* magnetizations isolated in the phyllites between ~650 and 680°C are a composite and that line fits used to estimate the direction of the K* overprint may be biased by a shallower yet unresolved component. The sites collected in phyllite were excluded from the final calculations. The combined mean of the Rajon intrusion and two sites in remagnetized Mesozoic strata is D=330.2°, I=43.4° (n=4 sites; Table 3). When compared to sites from the Pitiquito intrusion, declinations (farther west) and inclinations

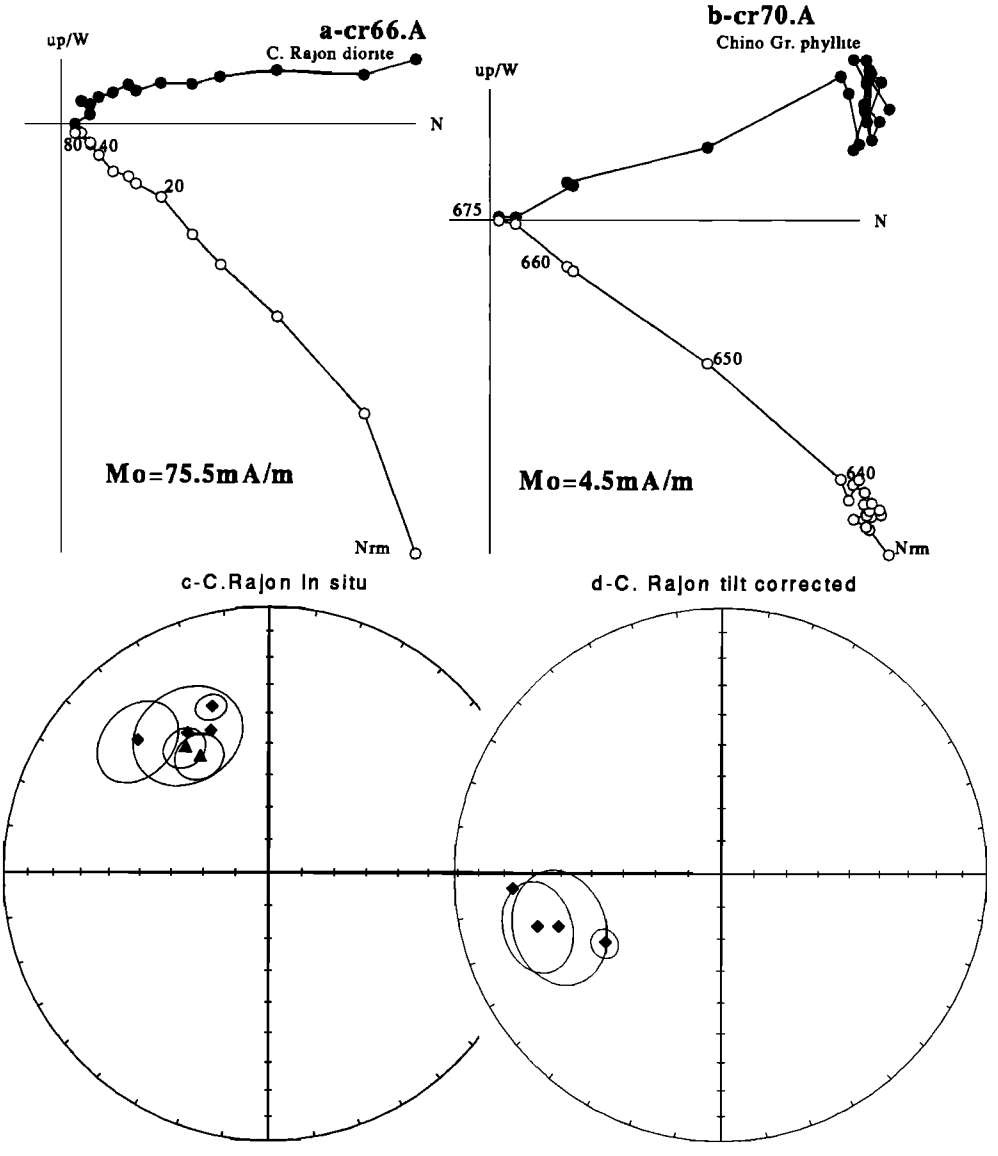


Figure 8. (a, b) Orthogonal demagnetization diagrams of samples collected at Cerro Rajon. Symbols are as in Figure 5. Figures 8c and 8d are equal-area projections of the site means and corresponding 95% cones of confidence for the characteristic magnetization in the Late Cretaceous Rajon intrusion and host Jurassic strata (diamonds).

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(shallower) at Cerro Rajon are consistent with small southeast-side downtilt similar to that of sites in Sierra la Vibora, but independent evidence for such tilt is lacking.

4.2.2.5. Cerro la Ventana: Lower Cretaceous (?) red beds assigned to the La Ventana Formation in the lower plate of the Sierra la Vibora thrust [*De Jong et al.*, 1988] are probably correlative with strata assigned to the Bisbee Group in Sierra el Chanate north of Caborca [*Jacques-Ayala*, 1995]. These rocks were sampled at Cerro La Ventana (Figure 4b). The red beds are characterized by near univectorial, shallow in situ north directed magnetizations of high coercivity and high laboratory unblocking temperature (>600°C; Figure 9a). Two samples from site lv76 yield reverse polarity magnetizations (Figure 9b). The red beds are overlain, in angular unconformity, by volcanic rocks of Miocene age (C. Jacques-Ayala, personal communication, 1997). The volcanic rocks, which dip gently to the northeast, have relatively straightforward demagnetization behavior, their NRM is nearly univectorial and their ChRM is north directed and of moderate to steep positive inclination (Figure 9c).

In situ site means of the La Ventana Formation are clearly distinct from those of the volcanic rocks, and they do not resemble a Cretaceous or younger paleofield direction (Figure 9d). After correcting for the tilt of overlying volcanic rocks and the dip of La Ventana strata, site means yield a mean of D=339.9°, I=47.9° (n=4 sites; Table 3), suggesting that acquisition of the ChRM predates folding. The ChRM of the red beds is therefore interpreted as a prefolding, near-primary magnetization clearly older than K* and K magnetizations. The volcanic rocks yield directions roughly consistent with the expected Tertiary direction (Figure 9e), but besides establishing the pre-Tertiary age of the magnetization in the red beds, no other inferences are possible because the small number of sites sampled is probably insufficient to average paleosecular variation.

4.2.2.6. Cerro Provedora and Ranch Bizani: Paleozoic carbonate rocks crop out in isolated hills in the valley west of Cerro Provedora (Figure 2, locality 6), near ranch Bizani (Figure 2, locality 6b) [*Brunner*, 1975]. Four sites were collected from a sequence of Upper Devonian, locally brecciated

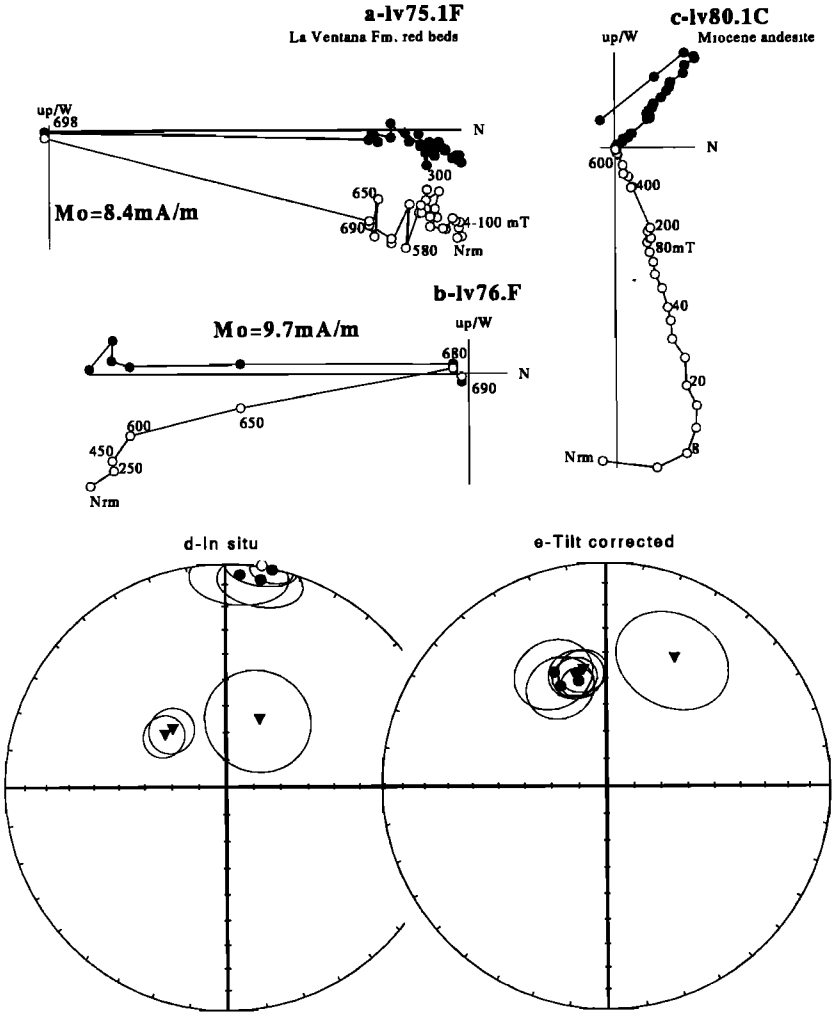


Figure 9. Orthogonal demagnetization diagrams of samples collected at Cerro la Ventana. Figures 9a and 9b are red beds of La Ventana Formation and Figure 9c is a Miocene andesite. Figures 9d and 9e are equal-area projections of the site means and corresponding 95% cones of confidence for the characteristic magnetization in La Ventana Formation (circles) and overlying volcanic rocks (inverted triangles).

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carbonate rocks (biz87-89), and from an overlying Mississippian (?) rhythmic alternation of carbonate mudstone and shale (site biz90). All sites collected in these rocks yield low-temperature (less than about 270°C) and low coercivity, north directed and steep positive magnetizations in the direction of the recent dipole field. In two of the sites, this is the only resolvable magnetization, and these two sites were excluded from further analysis. In site biz87, we resolved a moderately steep negative K* magnetization using great circle trajectories (Figures 10a and 10b). In site biz90, an intermediate temperature steep north-northeast directed magnetization is unblocked between 360 and 400°C, and a third magnetization, steep and west directed is defined by a linear trajectory to the origin between about 500° and 570°C (Figure 10c).

The tilt-corrected high-unblocking temperature magnetization in site biz90 is south directed and of shallow positive inclination (Figure 10h). It is thus similar to the characteristic north-northeast directed J* magnetization observed in Cerro Clemente and Cerro Gamuza but is defined by a single site and was excluded from the final calculations.

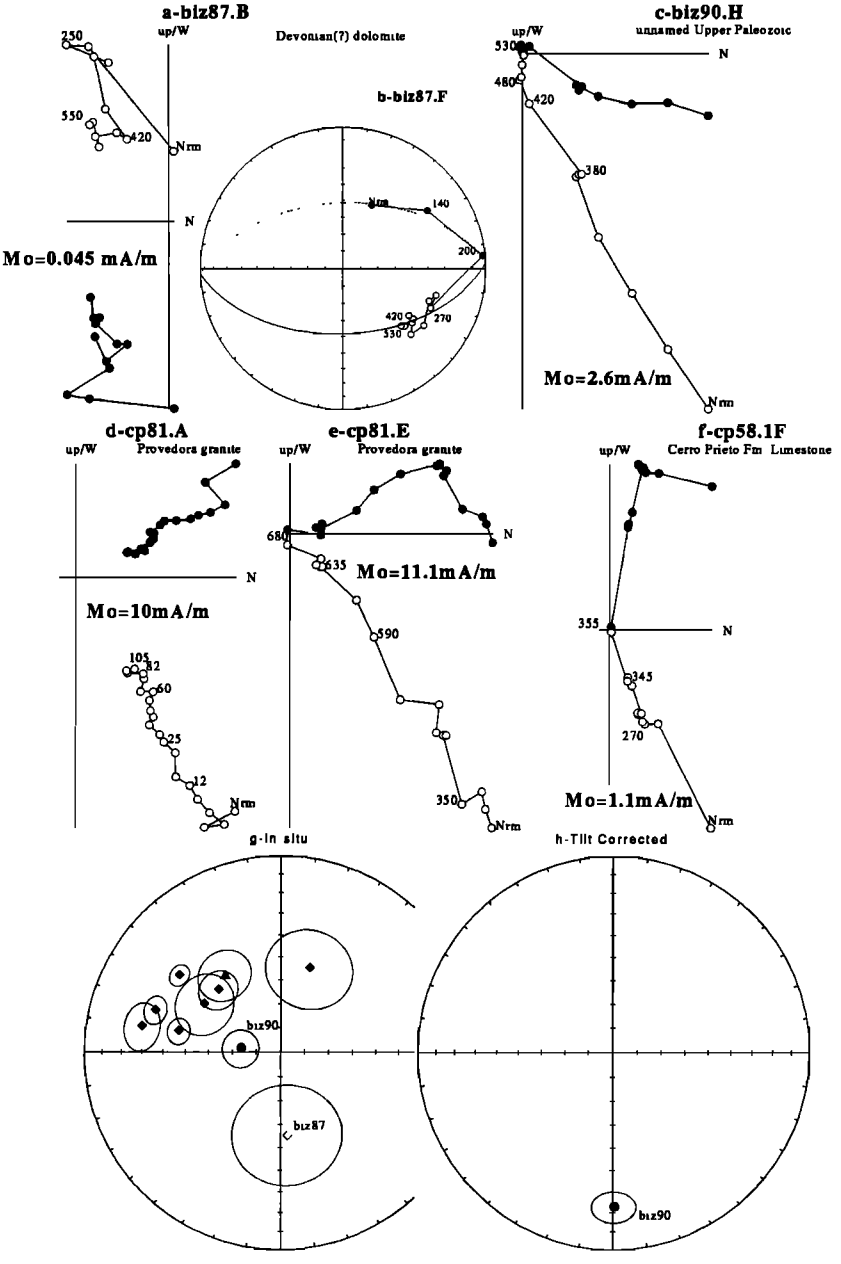


Figure 10. Orthogonal demagnetization diagrams of samples collected at (a, c) Bizani Ranch and (d-f) Cerro Provedora. Figure 10b shows an example of a great circle demagnetization trajectory of a sample from site biz87. Figures 10g and 10h are equal-area projections (in situ and tilt corrected, respectively) of sites means and corresponding 95% cones of confidence for the characteristic magnetization (circles), Cretaceous intrusion (triangle), and K* remagnetized miogeoclinal rocks (diamonds). Other symbols are as in Figure 5. See the text for further explanation.

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Cerro Provedora is a small, elongate, north-south trending range where steeply dipping to overturned uppermost Proterozoic and Cambrian strata are well-exposed (Figure 4c). The southern part of the range is a Cretaceous granite surrounded by skarn developed in the carbonate rocks. The Provedora granite yields a well-defined, northwest directed, and moderate to steep positive inclination K magnetization with maximum laboratory unblocking temperatures of 680°C (it is thus partially carried by hematite) overprinted by a prominent north directed steep positive magnetization (Figures 10d and 10e). The ChRM of Cambrian miogeoclinal carbonate rocks consists of a north directed and steep positive magnetization that is unblocked below 200°C, and a high-coercivity, steep to moderately shallow, northwest directed magnetization unblocked between 325° and 415°C (Figure 10f). We interpret this magnetization as a K* overprint of normal polarity.

Sites cp54 and cp55, on the east side of the range (Figure 4c), yield magnetizations that are similar to the site mean of the Provedora granite (Figure 10g). The remaining four sites, collected in the central and western part of the range, yield shallower west-northwest directed magnetizations. These are also interpreted as K* magnetizations because they yield a greater scatter of site means after tilt correction. Bedding attitudes on the east side of Cerro Provedora are near vertical. On the west side of the range, however, beds are overturned by as much as 120°. The change in bedding attitude (and the corresponding distribution of in situ site means) can be explained if the west side of the range was moderately tilted (east side down along a north trending fault) after folding and after intrusion of the Provedora granite. Other than the change in bedding attitude, there is no independent evidence of such tilt, but the general distribution of site means along a small circle (Figure 10g) and the presence of a strike valley along the central part of the range are consistent with this interpretation. We exclude the four sites on the west side of the range and calculate a mean for the Provedora granite and K* magnetizations of D=313.5°, I=52.2° (n=3 sites).

4.2.2.7. Sierra la Berruga, Cerros Calavera-Aquituni: Basement rocks are exposed on the northwest flank of Sierra la Berruga. This range consists of folded (east verging) Neoproterozoic and Cambrian strata [Damon et al., 1962; Eells, 1972]. Our sampling centered in the topographic gap between Cerro Calaveras and Cerro Aquituni in the Papalote, Cerro Prieto, Buelna, and Puerto Blanco Formations (Figure 4d). The ChRM of carbonate rocks is a moderate to steep positive inclination north-northeast directed magnetization that unblocks above 595°C. This K* magnetization is carried by hematite and is overprinted by a high coercivity and low unblocking temperature, north directed magnetization (Figure 11a) that may in part reside in goethite. Site cv31 contains K* magnetizations of both polarities, with gray dolomites invaded by hematitic veins containing a south directed and moderate to steep negative inclination magnetization of high unblocking temperature (Figure 11b).

Sites cv36 and cv37 are characterized by different magnetizations. The ChRM of these sites is east-southeast directed and of shallow inclination (Figure 11c). These are well-grouped at the site level, but a tilt test yields inconclusive results. Intermediate coercivities and unblocking temperatures suggest that it resides in magnetite. Two additional sites (cv38 and cv39) were collected from the eastern part of the range, east of a prominent north-northwest trending normal fault mapped along the entire length of Sierra la Berruga. The remanence of site cv38 consists of at least three interpretable magnetizations (Figure 11d). A north directed recent overprint is removed below 300°C, a shallow east-southeast directed magnetization unblocks between 300°C and 400°C, isolating a steep, northwest directed magnetization with maximum unblocking temperatures of 590°C. Site cv39 yields a magnetization of high coercivity that is slightly displaced from the cluster of northeast directed magnetizations from sites in the western side of the range (Figure 11e).

The northeast magnetizations of sites cv31 through cv36 (Table 2) are well-grouped at the site level and are found in sites with different structural correction. Tilt correction increases the scatter of site means, suggesting that remanence acquisition postdates folding (Figure 11e). The ratio of the in situ to the tilt-corrected precision parameter (k_{is}/k_{tc}) is 3.7, giving a negative fold test at the 99% probability level [*McElhinny*, 1964]. The steep northeast directed magnetizations are interpreted as K* overprints, but there are no Cretaceous intrusions exposed in this area. IRM acquisition curves of samples from Cerro Calavera are dominated by high-coercivity phases, show little evidence of magnetite, and in at least one instance show evidence of the presence of goethite (Figure 3a). The magnetizations of sites cv38 and cv39, northwest and northeast directed and steep positive, respectively, are likely to be K* overprints as well. The overall mean of K* magnetizations at Cerro Calavera is of D=17.8°, I=48.6° (n=8 sites; Table 3). In this locality there is no evidence of a south-southwest directed and shallow J* magnetization.

The geologic significance of the shallow east-southeast magnetizations at sites cv36 to cv38 is not well understood. Overall, two sites from other localities (bm40 and til22) yield (tilt corrected) east directed shallow magnetizations and one site (til29) gives a near antipodal magnetization.

4.2.2.8 Cerro Tilin-Pozos de Serna: In Cerro Pozos de Serna, Lower Jurassic strata of the Sierra de Santa Rosa Formation overlie Paleozoic carbonate rocks (S.G. Lucas et al., manuscript in preparation). A west dipping thrust on the west side of the range places Precambrian dolomites on Jurassic strata and a Cretaceous stock intrudes the upper and lower plate. Ammonites in Mesozoic strata had previously been identified as Late Jurassic [*Dowlen and Gastil*, 1981], but recent identifications indicate that the entire Mesozoic sequence is Liassic (S.G. Lucas et al., manuscript in preparation). Sites til20-til22 were collected in the west side in upper plate rocks of the El Arpa Formation. The remaining sites were collected in Neoproterozoic and Jurassic lower plate strata.

Demagnetization behavior of the diorite samples is unstable above 500°C (Figure 12a), but end-point directions, which are south directed and moderate to steep negative, yield a site mean of D=185.4°, I=-39.3°. A multivectorial magnetization is present in strata of the Sierra Santa Rosa Formation (Figures 12b and 12c). Shallow south-southwest directed J* magnetizations (or its antipode) are removed over a narrow range of unblocking temperatures between about 340°C and 400°C. A K* magnetization of high coercivity and high unblocking temperature (> 650°C) is south directed and of

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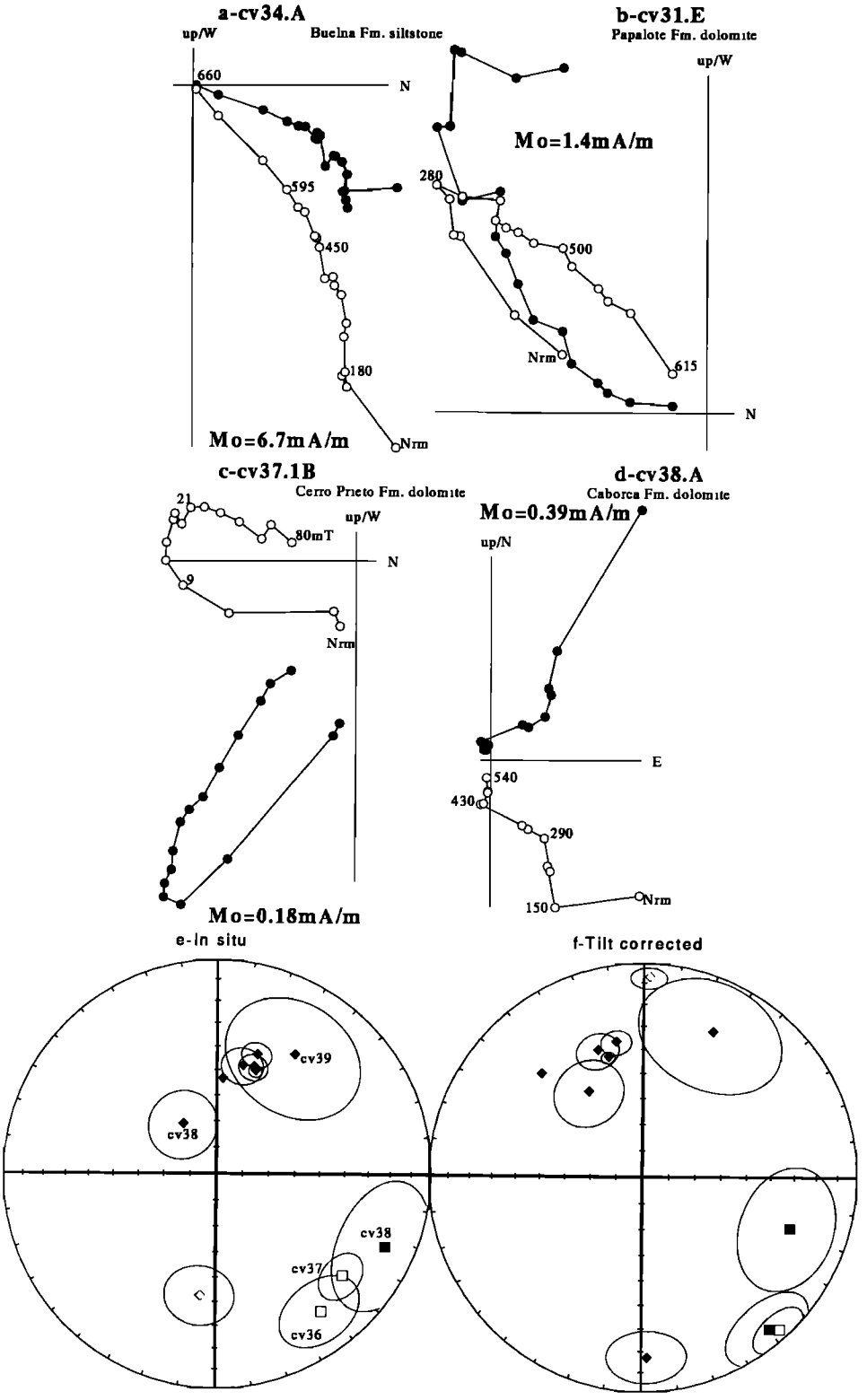


Figure 11. (a-d) Orthogonal demagnetization diagrams of samples collected at Cerro Calavera. Temperatures are in degrees Celsius. Figures 11e and 11f are equal-area projections (in situ and tilt corrected, respectively) of sites means and corresponding 95% cones of confidence for K* remagnetized miogeoclinal rocks (diamonds) and for other directions (squares). Other symbols are as in Figure 5. See the text for further explanation.

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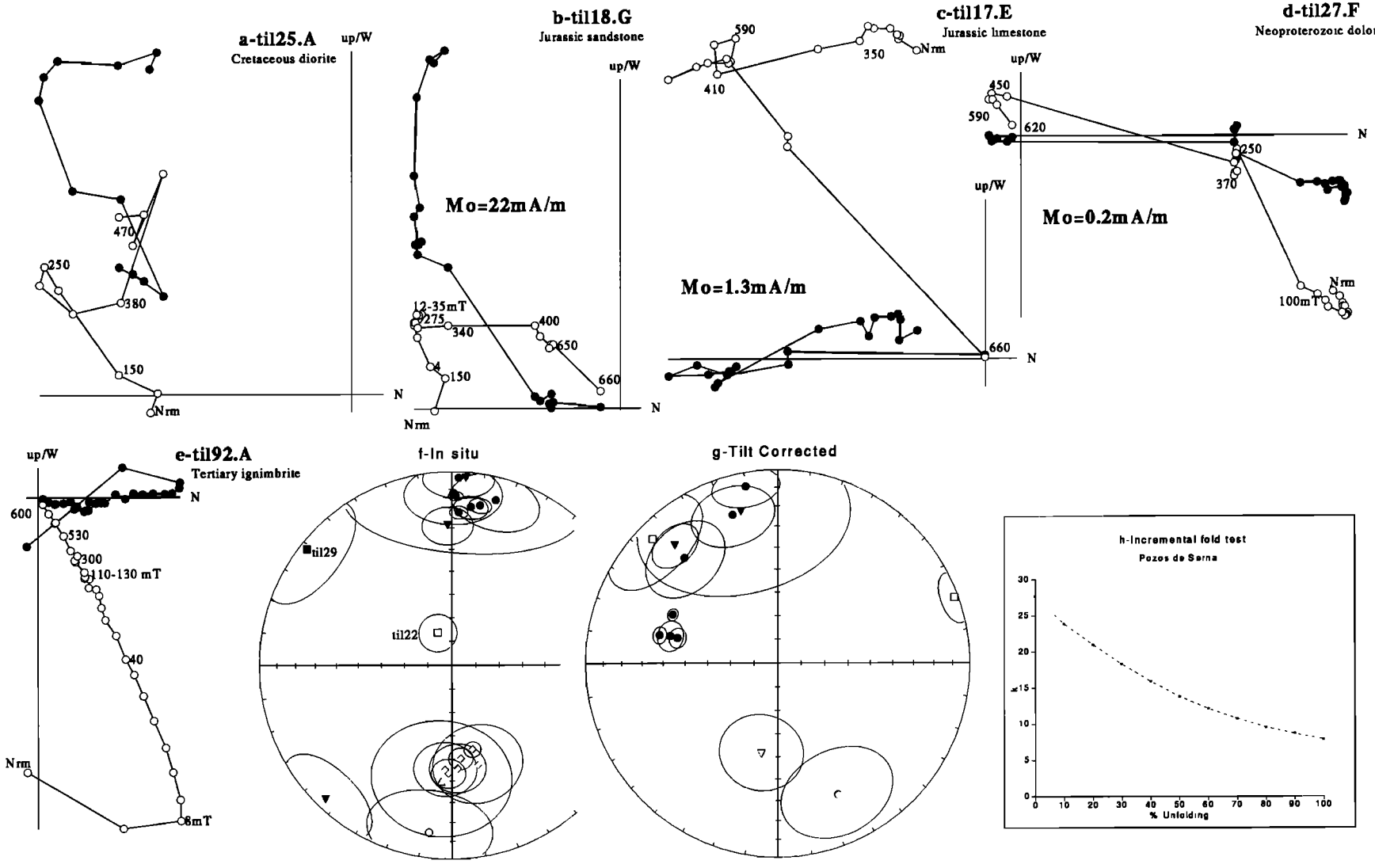


Figure 12. (a-e) Orthogonal demagnetization diagrams of samples collected at Cerro Pozos de Serna. Symbols are as in Figure 5. Figures 12f and 12g are equal area projections (in situ and tilt corrected) of sites means and corresponding 95% cones of confidence for the characteristic J* magnetization isolated in Precambrian strata (circles) and Jurassic strata (inverted triangles). Site means for Cretaceous intrusion (triangles) and K* remagnetized strata (diamonds) are also plotted in Figure 12f. Squares are used for other directions. Figure 12h shows results for an incremental fold test; notice that the maximum k is obtained in in situ coordinates.

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moderate to steep negative inclination. Two sites in Jurassic strata did not yield useful results (til19 and til24), and only the steep K* magnetization was observed in site til14.

The ChRM of Precambrian and Paleozoic strata is a high-coercivity (>100 mT) and intermediate-unblocking-temperature (< 450°C) shallow, east-northeast directed J* magnetization (or its antipode). The ChRM is overprinted by a high-coercivity, low-unblocking-temperature (< 250°C) magnetization in the direction of the recent dipole field (Figure 12d). A high-unblocking temperature (>600°C) K* magnetization of steep negative inclination and south directed is removed after the ChRM in two sites. Combined, K and K* magnetizations at Pozos de Serna yield a mean of D=175.1, I=-47.2° (n=7 sites; Table 3).

J* magnetizations in miogeoclinal strata are of reverse polarity at one site, and upper and lower plate rocks yield indistinguishable directions. In situ, the ChRMs of Jurassic (D=15.9°, I=9.5°; n=3) and miogeoclinal strata (D=6°, I=14.6°; n=8) are indistinguishable. A simple tilt correction increases dispersion of site means in both sets. A fold test using McFadden's [1990] V statistic for "limbs" with unequal k values yields a statistically significant, negative fold test (with 95% confidence), and the maximum value of the precision parameter is in in situ coordinates (Figures 12f-12h). The characteristic J* magnetization thus postdates folding.

Cenozoic (?) volcanic rocks crop out about 1 km southwest of Pozos de Serna. The general attitude of these rocks is mostly horizontal, but exposure is inadequate to make accurate bedding measurements. Also, exposure does not allow distinguishing between individual volcanic deposits. Site til93 is a volcaniclastic deposit and clearly overlies the other two sites. Site til93 and samples collected in the upper part of site til92 yield near univectorial magnetizations directed to the origin (Figure 12e). Moderate to high coercivity (>100 mT) and maximum unblocking temperatures of ~600° suggest that most of the remanence resides in magnetite. Magnetizations are north directed and moderate to steep positive, generally consistent with the expected late Tertiary paleofield direction. We interpret the directions in Tertiary volcanic rocks to indicate that large-magnitude Tertiary tilt or vertical axis rotation in the Pozos de Serna area are not significant.

Site til91 and the rest of the samples in til92 yield west directed and moderately shallow magnetizations. Demagnetization behavior of those samples is well defined and nearly identical to the behavior in other volcanic rocks. The anomalous directions (west and shallow) are not readily interpretable, but we note that Miocene ignimbrites in northeastern Baja California [*Lewis and Stock*, 1998] also yield anomalous directions that have been interpreted as transitional directions or as records of an excursion of the Earth's magnetic field. Similar ignimbrites have now been recognized in western Sonora (C. Lewis, personal communication 1998).

4.2.2.9. Sierra Santa Rosa: Sierra Santa Rosa (Figure 2, locality 9) exposes a sequence of Lower Jurassic sedimentary and volcanosedimentary rocks thrust on Precambrian crystalline basement [*Hardy*, 1981]. Neoproterozoic strata are limited to small, isolated klippes on both Jurassic strata and basement rocks. We sampled two sites in Proterozoic strata at Cerro Colorado (sr31 and sr32), and at a nearby locality (Cerro Delicias), we sampled Jurassic strata of the Sierra Santa Rosa Formation (sr33 to sr38). A far more extensive collection, sampled recently, is presently under study.

Magnetizations in Neoproterozoic rocks are in most cases relatively simple (Figure 13a). The ChRM is a south-southwest directed and shallow J* magnetization isolated after removing a prominent steep positive north directed magnetization. The ChRM unblocks over a narrow range of unblocking temperatures between 350 and 450°C. Some samples, however, display more complex magnetizations (Figure 13b). In site sr31, three components are evident after a spurious low coercivity magnetization is removed. These are a recent north directed overprint, the characteristic south-southwest directed J* magnetization, and a higher unblocking temperature K* magnetization that is north directed and moderate to steep in inclination. K* magnetizations of both polarities were also isolated in most of the sites collected from the Lower Jurassic Sierra de Santa Rosa Formation (Figure 13d). The high-temperature K* magnetizations yield an in situ mean of D=13.5°, I=42.3° (n=7 sites).

As in Pozos de Serna, Jurassic rocks contain north-northeast-directed shallow J* magnetizations. These magnetizations display distributed unblocking temperatures up to 650°C (Figure 13c). Site sr34 did not give useful data. The ChRM in sites sr37 and sr38 is problematic because directions are intermediate between the steep K* magnetization and the shallow magnetization of sites sr33, sr35, and sr36 (Table 2; Figure 13e). They seem to group better, however, with the steep K* magnetizations, and they were included as such in the final calculations. In situ magnetizations in Jurassic strata (D=28.4°, I=4.7°; n=3 sites; or D=20.7°, I=17.5°; if sr37 and sr38 are included) are indistinguishable from those in miogeoclinal strata (D=205.2°, I=7.1°: n=2 sites). Tilt correction results in a decrease in the k value of the overall mean for the five combined sites from 28.8 to 16.9 (Figures 13g and 13h). A fold test following *McFadden* [1990] is, however, inconclusive.

In the interior of Sierra de Santa Rosa, a mid-Tertiary conglomerate (the Baucarit Formation) is tilted some 25° to the west. After correcting K* directions for the tilt observed in Tertiary strata, these magnetizations yield a mean of D=349.9°, I=43.0°. Similarly, the overall mean of J* magnetizations after correction is D=20..9°, I=-11.0° (Table 3).

4.2.3. Localities north of the inferred trace of the Mojave-Sonora megashear. Details of the paleomagnetic results for sampling localities north of the inferred trace of the MSM (localities 10, 11a, and 11b; Figure 2) are described below.

4.2.3.1. Puerto el Alamo: The ranges immediately north of the city of Caborca are structurally complex. They consist of moderately to strongly deformed Mesozoic volcanic rocks of rhyolitic and andesitic composition, interbedded with sedimentary and volcaniclastic rocks. Other rocks include mafic to intermediate dikes and sills. Thrust imbrication and penetrative fabrics are locally present [*Corona*, 1979]. Liassic ammonites have been recovered from Sierra la Gloria (*Stanley and Gonzalez-Leon* [1995] cite *Jacques-Ayala* [1993] as having identified the ammonite *Vermiceras* sp.). Middle Jurassic U-Pb isotopic age determinations of about 165 Ma have been obtained from mafic dikes in Sierra Tajitos, east of Sierra la Gloria [*Stewart et al.*, 1986].

Jurassic rocks at Puerto el Alamo (locality 10; Figure 2) are

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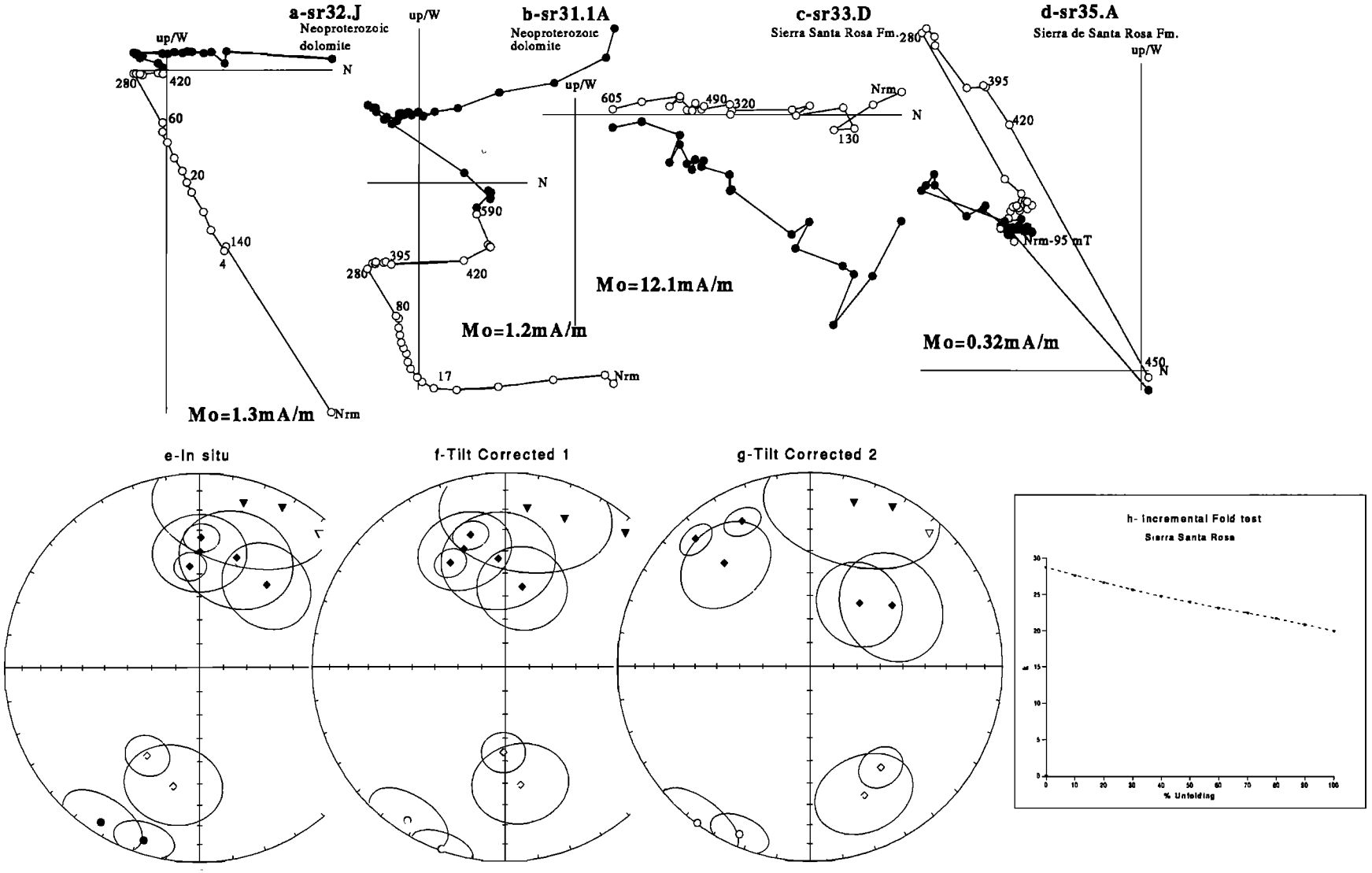


Figure 13. (a-d) Orthogonal demagnetization diagrams of samples collected at Sierra de Santa Rosa. Symbols are as in Figure 5. Figures 13e-13g are equal-area projections (in situ and tilt corrected) of sites means and corresponding 95% cones of confidence for the characteristic J* magnetization isolated in Precambrian strata (circles) and Jurassic strata (inverted triangles). Site means for K* remagnetized strata (diamonds) are also plotted in Figures 13e and 13f. Tilt correction 1 involves a simple tilt correction for the attitude of the Eocene Baucarit conglomerate. Figure 13h shows results for an incremental fold test; notice that the maximum k is obtained in in-situ coordinates.

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assigned to the Middle Jurassic La Maquina and the Upper Jurassic (?) Puerto el Alamo Formations [*Willard*, 1988]. Age assignments are based on broad regional correlations with volcanic sequences in Arizona and northwest Sonora [*Tosdal et al.*, 1990]. The volcanic rocks at Puerto el Alamo are overlain by Bisbee Group strata. East of Puerto el Alamo, at Sierra el Chanate, Jacques-Ayala et al. [1990] have recognized Jurassic volcanic and volcaniclastic rocks in contact with Upper Jurassic to Lower Cretaceous strata correlative with the Bisbee Group. These rocks are involved in Late Cretaceous contraction and low-grade metamorphism, which increases eastward.

At Puerto el Alamo we sampled roadcuts in a north-south transect along the south flank of an anticline that cores Sierra el Batamote [*Nourse*, 1995]. Our bedding data suggest that the anticline axis plunges gently to the west-northwest less than about 5°, not affecting significantly tilt corrections. The rocks collected consist of fine-grained hematitic sandstones, conglomerate and phyllites, interbedded with andesitic volcanic rocks, tuffs, and volcanic agglomerates. Site cal42 is stratigraphically the lowest in the section and site cal53 is the highest.

Fine-grained hematitic sandstones at Puerto el Alamo are characterized by essentially univectorial magnetizations with discrete unblocking temperatures above 650°C (Figures 14a). These magnetizations are north directed and have moderate negative inclinations. An andesite flow yields steep positive southwest directed magnetizations with unblocking temperatures above 650°C (Figure 14b). Two sites (cal46 and cal52) contain north directed and steep positive magnetizations (Figure 14c). Sites cal49 and cal51 are in an andesite and an overlying fine-grained sandstone. The ChRM in those sites is

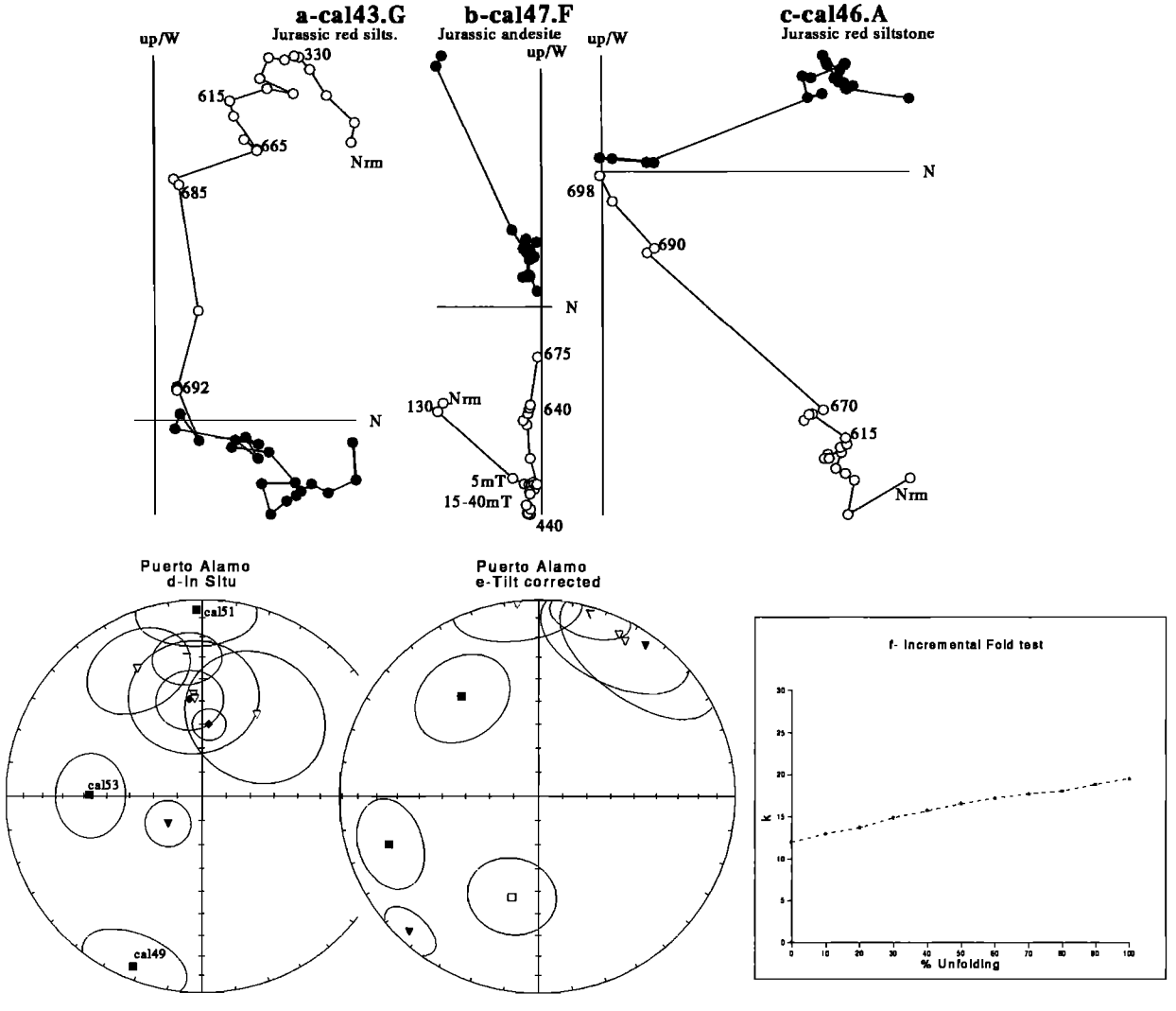


Figure 14. (a-c) Orthogonal demagnetization diagrams of samples collected at Puerto el Alamo. Figures 14d and 14e are equal-area projections (in situ and tilt corrected) of sites means and corresponding 95% cones of confidence for the characteristic magnetization isolated in Jurassic rocks north of the trace of the Mojave Sonora megashear (inverted triangles). The ChRMs at two sites are interpreted as K* magnetizations (diamonds). Squares are used for site means for other sites. Figure 14f shows results for an incremental fold test; notice that the maximum k is obtained in tilt-corrected coordinates.

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south or north directed, respectively, and shallow (Figure 14d).

After tilt correction, the group of (older) sites (cal42-cal45, cal47, and cal48) yields a cluster of north-northeast and shallow directions (Figure 14e). The site mean of the andesite flow becomes nearly antipodal (~160°) to the group of north-northeast directed sites. This magnetization is called the J magnetization. A fold test involving these sites is not statistically significant, but tilt correction improves the precision parameter k from an in situ value of 12 to a tilt-corrected value of 19.5 (Figure 14f). Below, we show that a regional fold test that includes these sites and sites from Cerro Basura is statistically significant at the 95% probability level. Also, a positive fold test is consistent with the fact that in situ magnetizations yield inclinations that do not coincide with the Jurassic or younger expected paleolatitudes for southwest North America. Six sites yield a locality mean of D=24.0°, I=-4.4° (Table 3). *Harrar* [1989] reported paleomagnetic data for Mesozoic strata in Sierra el Chanate and Sierra el Batamote but concluded that magnetizations from rocks at Puerto del Alamo are secondary, giving a pole position near the present-day pole. Although steep north directed magnetizations are present, our observations are not fully consistent with that interpretation.

We interpret the north directed and moderate to steep positive magnetizations at sites cal46 and cal52 as K* magnetizations; their combined mean is D=358.3°, I=54.7°. Two of the remaining sites yield shallow magnetizations (cal49 and cal51), and one site (cal53) yields a west directed magnetization. In tilt-corrected coordinates these sites yield inclinations of 36° to 44°, significantly steeper than the inclinations observed in the lower part of the stratigraphic section (Figure 14e). A bedding-parallel shear zone separates the group of sites with low (tilt corrected) inclinations from the group of sites with moderate inclinations; also, south of this shear zone there is a greater variation in bedding attitudes and greater density of faults. We suspect that the ChRM of sites cal49, cal51 and cal53, which again, are stratigraphically higher, was acquired before folding and that the magnetizations in these sites are significantly younger than those in sites cal42 to cal48. Additional study is required to support this interpretation. We opted to exclude sites cal49, cal51, and cal53 from further consideration.

4.2.3.2 Cerro Basura and Sierra la Gloria: At Cerro Basura (locality 11a; Figure 2) we sampled gray litharenites and siltstones, as well as mafic to intermediate volcanic rocks (bas54-bas58; Table 1). These rocks have been assigned a Early to Middle Jurassic age by *Corona* [1979]. In the foothills of Sierra la Gloria (locality 11b; Figure 2) we sampled river bed exposures of medium- to coarse-grained hematitic sandstones interbedded with red mudstones and conglomerate (lv70-lv73). We infer these rocks to be Early Cretaceous in age because deformation is less intense than in the Jurassic strata; no fossils have been reported from these rocks.

The rocks from Cerro Basura have relatively simple behavior during demagnetization. Alternating field and thermal demagnetization both isolate a dual-polarity, shallow (north or south directed) J magnetization that resides in a cubic phase such as magnetite (Figure 15a). In either in situ or tilt-corrected coordinates the J magnetization is shallow. One site collected in coarse-grained sandstones at Cerro Basura (bas57) yields a steep negative, southeast directed magnetization interpreted as a K* overprint (Figure 15b). The precision parameter of four site means is of a higher value in in situ (Figure 15d) coordinates (28.4) than in tilt corrected coordinates (21.3), with a maximum of 35 at 40% unfolding, but the change is not statistically significant. In tilt-corrected coordinates (Figure 15e), however, the mean (D=1.2°, I=15.6°, n=4 sites) agrees better with results from Puerto el Alamo, suggesting that the magnetization predates folding. In fact, a regional fold test that combines the four sites from Cerro Basura with six sites from Puerto el Alamo yields a statistically significant fold test, with the maximum k value observed at 100% unfolding (Figure 15f). Using *McElhinny*'s [1964] criteria, the ratio of the precision parameter in tilt-corrected to that of in situ coordinates (k_{tc}/k_{is}) is 3.05 which for n=10 sites indicates a positive fold test at the 95% probability level.

The sequence of hematitic sandstone, mudstone, and conglomerate in the foothills south of Sierra la Gloria is characterized by an exclusively south-directed, steep negative magnetization of high coercivity and distributed unblocking temperatures between 200 and 620°C (Figure 15c). The rocks dip uniformly to the northeast, and a fold test is not possible. The tilt-corrected ChRM is south-southwest and shallow, which is inconsistent with expected Early Cretaceous directions. The in situ direction is interpreted as a K* overprint. Cretaceous intrusions are exposed a few kilometers southeast of this locality in isolated outcrops. Combined with the direction observed in site bas57, we compute a locality mean for K* magnetizations of D=167.1°, I=-54.9° (Table 3).

4.3. Interpretation

4.3.1 Cretaceous magnetizations. Late Cretaceous intrusive rocks collected from widely separated localities (Cerro del Arpa, Cerro la Provedora, Cerro Pitiquito, Cerro Rajon, and Cerro Pozos de Serna; Figure 2), yield dual-polarity magnetizations that collectively pass a reversal test, with a mean of D=341.4°, I=52.3° (n=10 sites; α_{95}=7.9°; k=38.3). The Pitiquito and Aibo intrusions yield K-Ar isotopic age determinations of ~80 Ma (biotite and amphibole [*De Jong et al.*, 1988]) and ~85 Ma (biotite [*Damon et al.*, 1962]), respectively. As noted earlier, miogeoclinal rocks in the vicinity of Cretaceous plutons have K* magnetizations indistinguishable from those in the intrusions. The K* magnetizations have been isolated from six additional localities where there is no surface evidence of Cretaceous magmatism or where Cretaceous intrusions were not sampled during this study (Sierra de Santa Rosa, Cerro Calavera, Cerro Clemente, Bizani, Puerto el Alamo, and Cerro Basura). Their interpretation as K* magnetizations is based on the similarity between directions obtained from these localities and those in Cretaceous rocks and the fact that in all these localities K* magnetizations are carried by a phase of high-coercivity and high unblocking temperature, most likely hematite. We suspect that the Cerro Provedora (D=313.5°, I=52.2°; n=3 sites), Cerro Rajon (D=330.2°, I=43.4°; n=4 sites), and Cerro Calavera (D=13.6°, I=48.6°; n=8 sites) localities may have experienced modest tilting associated with Tertiary extension. These localities yield means with shallower directions roughly distributed along a small circle that includes the overall mean and is semi

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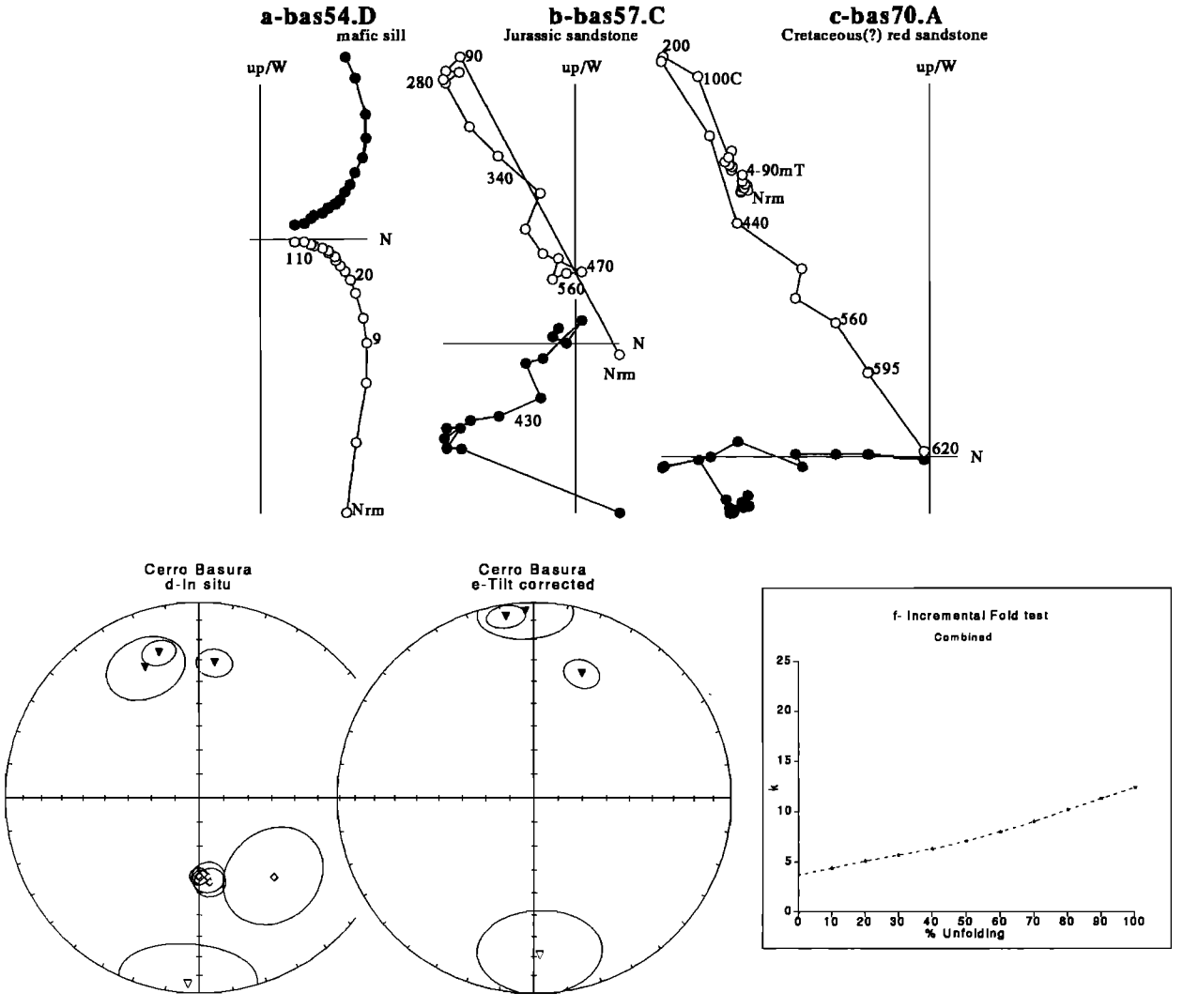


Figure 15. Orthogonal demagnetization diagrams of samples collected at (a, b) Cerro Basura and (c) the foothills of Sierra la Gloria. Figures 15d and 15e are equal-area projections (in situ and tilt corrected) of sites means and corresponding 95% cones of confidence for the characteristic magnetization isolated in Jurassic rocks north of the trace of the Mojave Sonora megashear (inverted triangles). K* magnetizations are plotted as diamonds. (f) Results for an incremental fold test combining data from Cerro Basura and Puerto Alamo; notice that the maximum k is obtained in tilt corrected coordinates. Symbols are as in Figure 5.

perpendicular to the regional trend of Tertiary extension (north-northwest). Lacking independent evidence of tilt we have no reason to exclude those localities from the overall mean.

The population of 24 normal and 38 reverse polarity site means passes a reversal test (Figure 16a). The angle between the means of normal and reverse directions is 3.5°, and it is almost entirely a difference in declination. The test is classified as B according to *McFadden and McElhinny* [1990]. This positive reversal test suggests that K and K* magnetizations are not significantly biased by incomplete removal of a young north-directed overprint which would be manifest mainly as differences in inclination.

If the data from all 11 localities are combined at either the site or the locality level, indistinguishable mean directions result (Table 3; Figure 16b). The mean of the 11 localities is D=350.8°, I=51.0° (α_{95}=6.6°; and k=48.2), a grand Late Cretaceous mean calculated giving unit weight to each site mean is D=353.7°, I=50.3° (n=62 sites; Table 3). These results are based on data from rocks with magnetizations that lack reference to the paleohorizontal, and an important concern is the possibility of local tilt associated with Tertiary extension. We have assumed that the principal source of directional dispersion is tectonic. Therefore the best estimate of the Late Cretaceous field direction in the Caborca region is the average of locality means, which yields a paleomagnetic pole at 82°N, 169°E (for a central locality at 30.5°N, 112°W). Latitudinal displacement (3.3°±2.8°, northward) and relative rotation (9.1°±8.7°, clockwise), inferred from comparison with a direction calculated for a reference mean pole at 73°N 190°E

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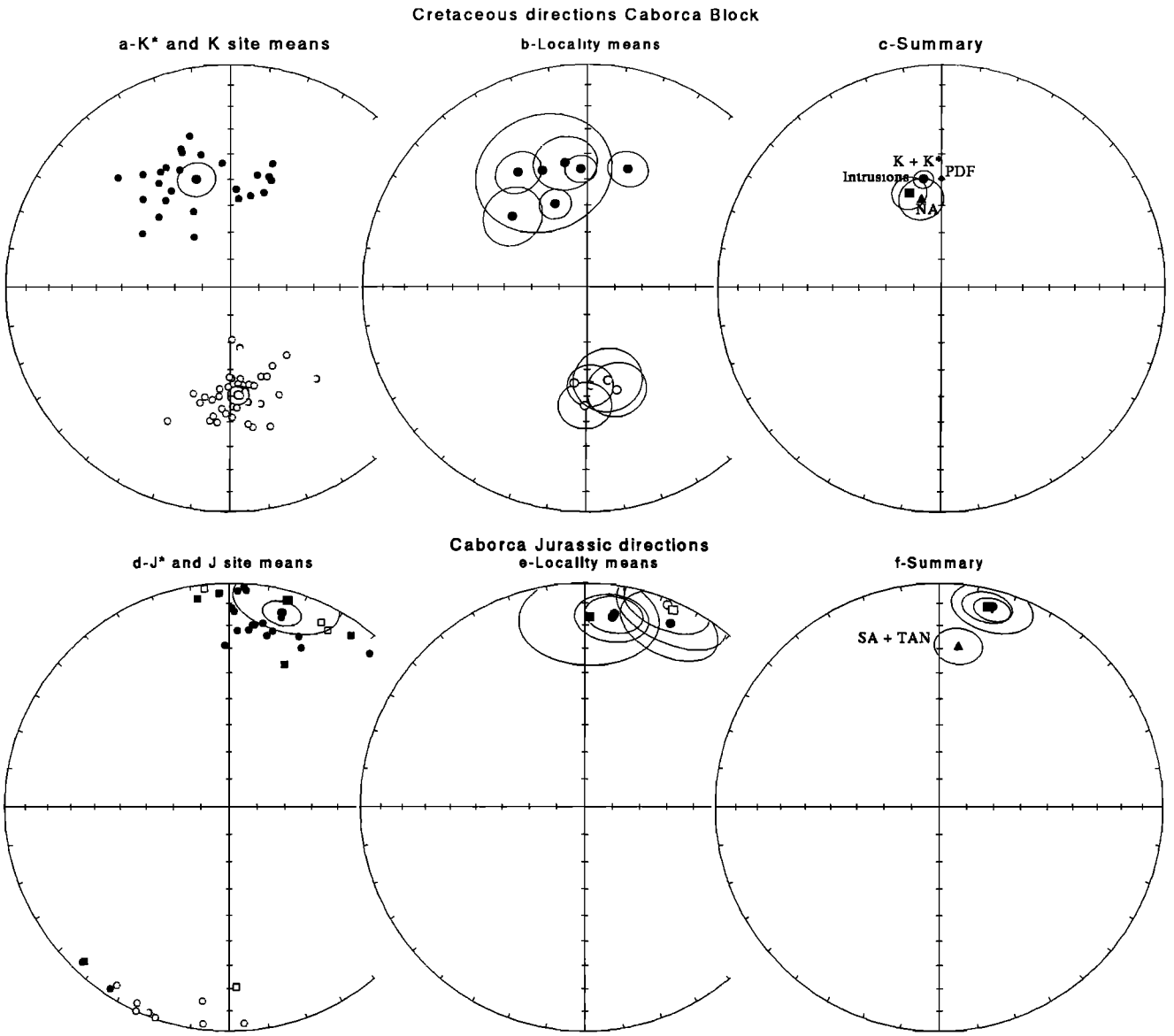


Figure 16. Summary of paleomagnetic data for the Caborca area. (a) Site means of K and K* magnetization with the associated mean of normal and reverse polarity sites (large symbol with 95% confidence region). (b) Mean locality means of K and K* magnetization (circles). (c) Overall site mean (circle labeled K+K*), mean of Cretaceous intrusions (square), and expected Late Cretaceous (68-90 Ma) direction (triangle labeled NA). The Quaternary dipole field is plotted as a diamond. (d) Site means of J and J* magnetization in rocks north (squares) and south (circles) of the MSM, compared to their respective mean (larger symbol with confidence region). (e) Comparison of locality means for rocks north (squares) and south (circles) of the MSM. (f) Overall site mean (n=38 sites), selected site mean (n=24 sites), and locality mean (N=6) for J and J* magnetizations combined. Also plotted is the mean for the Antimonio Formation (Sierra el Alamo and Los Tanques localities), south of the MSM after *Molina Garza and Geissman* [1996a].

(using eight selected pole positions for North America, in the range from 90 to 65 Ma, obtained from the global paleomagnetic database of *McElhinny and Lock* [1996]), indicate small to negligible post-Late Cretaceous northward motion and a modest clockwise rotation of the Caborca block with respect to the craton. The selection of a Late Cretaceous reference pole is critical in the interpretation. For example, a comparison based on the expected direction from the ~ 80 Ma Late Cretaceous pole of *Diehl* [1991] (80.3°N 189.5°E) yields estimates of latitudinal displacement and clockwise rotation of 1.8°±4.1° and 1.1°±11.2°, respectively, indicating negligible latitudinal displacement or rotation.

Overall, the paleomagnetic data from the broad area we sampled imply that effects of post-remanence acquisition tilt or rotation are small and that the amount of both rotation and latitudinal shift derived from comparison with appropriate reference directions is also small. These rotation and displacement estimates should be carefully scrutinized, espe-

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cially in the context that results for the intrusions alone indicate statistically insignificant rotation and latitudinal displacement. The overall mean of K* magnetizations is a few degrees shallower and more north-directed than the mean for the intrusions (Figure 16c). In some sites with complex multivectorial magnetizations, this could be attributed to incomplete separation of the steep K* magnetizations and the characteristic, south-southwest directed magnetization. As suggested earlier, in other sites, shallow inclinations may be caused by incomplete removal of a prominent recent overprint. Accordingly, we would expect that the best defined site means should yield a less biased estimate. The overall mean of 47 selected sites with $n>2$ samples, $k>10$, and $\alpha_{95}<20°$ (D=351°, I=50.6°), and the corresponding mean of 10 localities (D=348.1°, I=50.7°) indicate overall similar estimates of latitudinal displacement and rotation than those derived from the complete data set. The mean of 10 localities yields latitudinal displacement and clockwise rotation estimates of 3.4°±2.9° and 6.5°±8.7°, respectively.

4.3.2. Cretaceous rocks, (Bisbee Group) Cerro la Ventana. The Cerro la Ventana sites in Lower Cretaceous rocks were collected in steeply southwest dipping strata unconformably overlain by Tertiary volcanic rocks. In in situ coordinates the ChRM does not resemble the Cretaceous or younger paleofield direction, but in tilt-corrected coordinates the ChRM is northwest directed (D=339.9°, I=47.9°; α_{95}=5.6°; k=242.0). The thickness of the interval sampled (~100 m), the magnitude of the circular standard deviation of site means, and the presence of both polarities are all consistent with secular variation being adequately averaged. The tilt corrected mean direction yields a paleomagnetic pole (72.5°N, 157.4°E) that falls along the latest Jurassic-Early Cretaceous segment of the North American apparent polar wander path (APWP), as defined by the upper Morrison Formation and Monteregian Hills reference poles [*McElhinny and Lock*, 1996]. Tectonic interpretations for this result are tentative. Although magnetizations are well defined, the results are based on a small number of sites from one locality. These data do not support significant post-Early Cretaceous rotation of the sampling area.

We note that the mean direction for La Ventana Formation rocks is similar to the tilt-corrected low-unblocking temperature magnetization in basal Precambrian strata (cg26 and cg27) and a Precambrian dike at Cerro el Arpa (ab25). Although coincidental, these Cerro el Arpa magnetizations may have been acquired prior to folding (thus suggesting that the shallow ChRM at this locality must also predate folding). Other interpretations are possible, however. The NRMs in the Cerro el Arpa rocks (sites cg26 and cg27) are complex, multivectorial magnetizations, and the low-unblocking-temperature component is defined over a very narrow range of unblocking temperatures. This may be composite and therefore of geologically meaningless origin. Alternatively, this may represent a postfolding Cretaceous secondary remanence that was acquired prior to Cretaceous magmatism and may indicate a modest amount of local, pre-intrusion clockwise rotation or tilt. This explanation could account for the small difference between data obtained from the Cretaceous intrusion and data from remagnetized miogeoclinal strata at this locality. As noted earlier, K* magnetizations at sites cg26 and cg27 have shallower inclinations, averaging about 40°, and their declinations are west of those in intrusive rocks.

4.3.3. Jurassic volcanic arc rocks north of the trace of the MSM. Jurassic volcanic and volcaniclastic rocks yield prefolding shallow, east-northeast directed J magnetizations; both polarities were observed (Figure 16d). The angle between the means of normal and reverse polarity magnetizations is 9°. The means are statistically indistinguishable; thus these data pass a reversal test with a C classification according to *McFadden and McElhinny* [1990]. This implies that the observed directions are not significantly biased by incomplete removal of a recent overprint. The overall mean combining the Puerto Alamo and Cerro Basura sites is D=15.0°, I=4.0° (α_{95}=14.3°; k=12.4; N=10 sites), and it yields a paleopole at 58°N, 39°E, which is discordant in a clockwise sense with respect to the early Mesozoic segment of the North American APWP.

The interpreted Jurassic age of rocks sampled north of Caborca is based on the presence of late Early Jurassic ammonites at Sierra la Gloria and isotopic U-Pb determinations of ~175 to 165 Ma for similar sequences north and east of Sierra la Gloria [*Stewart et al.*, 1986]. The age of this magnetization is most likely bracketed between about 190 and 160 Ma or, roughly, late Early to Middle Jurassic. The mean inclination indicates a near-equatorial paleolatitude.

The best reference pole for the late Early Jurassic North American APWP is from the Kayenta Formation of Utah and Arizona, on the Colorado plateau. The mean of three poles for the Sinemurian-Pliensbachian Kayenta Formation (~195 Ma) predicts inclinations of about 4° for a locality near Puerto el Alamo, assuming stability with respect to the craton. Younger poles, such as from the Callovian-Oxfordian (~155 Ma) Summerville Formation [*Bazard and Butler*, 1992], predict inclinations of about 25°. To the limit of paleomagnetic resolution, observed inclinations (+4.0°) may thus indicate (depending of the exact age of the magnetization) negligible to modest (~1000 km) northward latitudinal displacement of the sampling region relative to North America. Declinations indicate a significant clockwise rotation ranging from ~12° (195 Ma) to ~50° (160 Ma). We note that a comparison based on Middle Jurassic North American reference poles at high latitudes [*Van Fossen and Kent*, 1990] indicates a significantly greater northward displacement (~2000 km) of the sampling region with respect to the craton.

4.3.4. Characteristic magnetization in Neoproterozoic miogeoclinal strata and Jurassic strata south of the trace of the MSM. A shallow south-southwest directed J* magnetization (or its antipode) is well-defined in rocks from four localities south of the inferred trace of the Mojave Sonora megashear (Figure 16d). It may also be common to two other localities (Bizani and Sierra la Vibora; Table 3), but more study is necessary for those areas. In Pozos de Serna and Sierra de Santa Rosa, the ChRM of Lower Jurassic strata is indistinguishable from the ChRM of Neoproterozoic miogeoclinal rocks. The magnetizations in Jurassic and miogeoclinal strata are therefore interpreted to be of the same age. Also, in these two localities the ChRM postdates folding, an unexpected result because deformation in the Caborca region had been inferred to be of Middle to Late Cretaceous age [*Jacques-Ayala et al.*, 1990; *Hardy*, 1981]. We note that the

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contact between the Sierra de Santa Rosa Formation and Neoproterozoic strata at Sierra Santa Rosa is a thrust, but at Cerro Tilin, in the Pozos de Serna area, the contact is depositional.

The shallow, near-equatorial paleolatitudes inferred from the mean inclinations observed in both Pozos de Serna (I=13.3°) and Sierra Santa Rosa (I=11.0°) are not consistent with a Middle to Late Cretaceous age of folding. If miogeoclinal Neoproterozoic and Jurassic Sierra de Santa Rosa Formation strata had been remagnetized in Late Cretaceous time, the shallow inclinations would require remagnetization at considerably lower latitudes. Alternatively, remagnetization could have occurred, after folding, near their present location with respect to North America, but the rocks were subsequently tilted by nearly 60° about an east-southeast trending horizontal axis. The second possibility is totally inconsistent with available geologic data. East-southeast trending structures are not present in this area, and fold geometries are not suggestive of possible apparent rotations [*MacDonald*, 1980]; rather, the general strike of sections collected is roughly perpendicular to this orientation. A more complex possibility, such as Late Cretaceous remagnetization at intermediate latitudes and subsequent moderate tilt, is also improbable. Strikes of the miogeoclinal and Jurassic rocks range between ~160° at Cerro Pozos de Serna and ~220° at Cerro Gamuza. Strikes are uniform at a range scale, suggesting that fold axes are horizontal. Tilting of a north-northwest directed and steep Cretaceous magnetization about such strikes would largely affect declination. Furthermore, a near identical tilt should have affected all four localities after folding and remagnetization, prior to intrusion of Late Cretaceous granitoids and acquisition of K* magnetizations.

Late Cretaceous remagnetization at near-equatorial latitudes is equally unlikely. First, the shallow ChRM is of both polarities. Therefore acquisition of a secondary magnetization should have occurred after the Cretaceous "long normal" superchron, which ended at about 84 Ma. Problematically, Cretaceous intrusions with primary magnetizations yield concordant paleomagnetic data of nearly the same age. In addition, Lower Cretaceous rocks of La Ventana Formation with prefolding magnetizations fail to indicate the northward displacement that remagnetization at equatorial latitudes would imply.

On the basis of the considerations presented above, we assert that our conclusion of remagnetization at an earlier time, in the Jurassic, is inescapable. This conclusion does not imply that Late Cretaceous compressional deformation did not affect regions of the Caborca block. Cretaceous deformation in the ranges north of Caborca is unquestionable [*Jacques-Ayala et al.*, 1990], and at Sierra la Vibora, Cretaceous deformation is also obvious [*De Jong et al.*, 1988]. One possibility is that Cretaceous deformation is less intense in the ranges south and west of a front of intense Cretaceous deformation delineated by the approximate trace of the inferred MSM. Older compressional deformation is probably responsible for most of the structures observed in the ranges south and west of Caborca. Cretaceous strata in those areas are absent; thus the inferred timing of deformation has been extrapolated from observations at Sierra la Vibora and other areas to the east. Middle or Late Jurassic deformation in western Sonora has been recently proposed by *Molina Garza and Geissman* [1996a] for rocks of the Antimonio terrane, based on paleomagnetic data and arguments similar to those presented here.

A fold test at Cerro Clemente and at Cerro el Arpa cannot be performed because strata are uniform in orientation at both localities. A regional tilt test (including Pozos de Serna and Sierra Santa Rosa) suggests that J* magnetizations postdate folding in all four localities south of Caborca. An overall mean which includes the in situ locality means from Sierra Santa Rosa and Pozos de Serna and the tilt-corrected locality means of Cerro Clemente and Cerro el Arpa (D=18.9°, I=-2.7°), is similar to the mean of in situ data (D=16.0°, I=9.6°), but the mean direction precision (k) is reduced from 44.9 to 15.3, yielding a negative fold test [*McElhinny*, 1964]. The k value is maximized in in situ coordinates.

At the 95% confidence level, the overall mean of sites south of the inferred trace of the MSM (D=15.0°, I=10.0°; α_{95}=5.8°; k=23.0; N=28 sites) is statistically indistinguishable and only 6° away from the mean of J magnetizations isolated in rocks of the Jurassic Cordilleran arc (Figure 16f). For the distribution of site means, a Monte Carlo simulation following *McFadden and McElhinny* [1990] indicates that the probability of exceeding 6° of angular distance is greater than about 0.6. We interpret the shallow north-northeast (or south-southwest) J* magnetization in miogeoclinal strata to be a Jurassic secondary magnetization, of about the same age as J magnetizations in rocks of the Cordilleran volcanic arc.

J* magnetizations have been observed at localities where structural relationships between basement rocks and miogeocline strata are widely different. In the ranges south of the megashear, J* magnetizations postdate folding, although in the ranges to the north, J magnetizations pass a fold test. The youngest strata south of Caborca that carry secondary J* magnetization are Pliensbachian in age, at Pozos de Serna. The older age limit for the J* magnetization is Toarcian (latest Early Jurassic). Because J* magnetizations are of dual polarity, they must predate the long normal Cretaceous superchron. The younger age limit is thus Aptian. The interval bracketing the timing of acquisition of J* magnetizations (about 190 to about 120 Ma) is in good agreement with the best estimate of the age of volcanic and volcaniclastic rocks in the ranges north of Caborca (190 to 160 Ma). Magnetizations of similar east-northeast and shallow directions have been isolated in lower Mesozoic strata in Sierra del Alamo Muerto and Barra los Tanques, about 40 km west of the localities we report here [*Molina Garza and Geissman*, 1996a], and Sierra Santa Teresa, near Hermosillo [*Molina Garza and Geissman*, 1996b]. It thus appears that an episode of Jurassic deformation and regional remagnetization is widespread in western Sonora.

In summary, we believe that we have provided ample evidence for the presence of two secondary magnetizations in Neoproterozoic miogeoclinal and Lower Jurassic arc-derived strata south of the inferred trace of the MSM. Based on the presence of dual-polarity magnetizations of shallow inclination, which are significantly different from magnetizations in Late Cretaceous intrusions and Lower Cretaceous red beds, remanence acquisition of the older J* magnetizations is reasonably bracketed between about 120 and about 190 Ma. This interpretation implies that most of the compressional deformation observed in the ranges south and west of Caborca is

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not "Laramide" or "Sevier" in age. With a 95% confidence, these J* magnetizations are indistinguishable from possible primary J magnetizations in Jurassic volcanic and volcaniclastic rocks north of the inferred trace of the MSM.

5. Tectonic Implications

5.1. Cretaceous deformation

The paleomagnetically determined modest rotation and latitudinal displacement of the Caborca region can be related to known aspects of the Cretaceous evolution of this area. A Late Cretaceous event involving ductile deformation is recorded in northwest Sonora, from the Sonoyta area, through the Altar desert, to the Santa Ana area, a belt that coincides with the trace of the MSM (Figure 1b). In the Altar-Santa Ana area, *Damon et al.* [1962] interpreted the Altar Schist as a product of late Mesozoic to early Tertiary metamorphism. More recently, K-Ar dates of about 60 Ma have been obtained from Altar Schist rocks [*Hayama et al.*, 1984]. The Altar Schist consists of intermediate- to low-grade metamorphic rocks, described at Sierra El Batamote by *Jacques-Ayala and De Jong* [1996]. The protoliths of the Altar Schist are interpreted to be volcanic and volcaniclastic rocks of the Jurassic Cordilleran arc and siliciclastic rocks of the Bisbee Group.

The Altar Schist has been interpreted as the southeastward extension of the Orocopia Schist, exposed in southeast California [*Jacques-Ayala and De Jong*, 1996]. The Orocopia Schist is a late Mesozoic- early Tertiary subduction complex juxtaposed against Precambrian through Mesozoic continental basement along the Vincent-Chocolate Mountains fault zone and similar structures [*Jacobson and Dowson*, 1995]. Paleomagnetic data for Cretaceous rocks of the Caborca terrane, at a regional scale, may be interpreted to indicate that the kinematics of Late Cretaceous deformation and metamorphism forming the Altar Schist involved small clockwise rotation (<~10°) and possibly some northward motion (<~300 km). The fact that Lower Cretaceous Bisbee Group strata are exposed on both sides of the outcrop belt of the Altar Schist is consistent with the hypothesis that northward displacement, permissible by observed inclination anomalies of less than about 5°, is indeed less than a few hundred kilometers.

5.2. The Mojave-Sonora megashear

Paleomagnetic data provide independent evidence that is inconsistent with the predictions of the Mojave-Sonora megashear model. Assuming J* magnetizations are secondary, their age is bracketed between latest Early Jurassic and Early Cretaceous, and these magnetizations bear on kinematic models proposed for the evolution of Sonora. Concerns regarding structural attitudes at the time of remanence acquisition of the rocks and the possibility of local, vertical axis rotation are reasonable; they have been adequately addressed above. The small scatter of locality means spread over an area of more than 5000 km^2 suggests, as it did for Cretaceous rocks, that deformation involving rotation or tilt after acquisition of the characteristic remanence is negligible. Importantly, we observe magnetizations of Jurassic age on both sides of the presumed trace of the megashear, and these are statistically indistinguishable (Figure 16f). This suggests that a Jurassic crustal discontinuity, along which major displacement occurred, does not exist at the proposed location of the MSM. This conclusion, we note, is independent of the configuration of the Jurassic North American apparent polar wander path chosen as a reference for our paleomagnetic data. It should be noted that J* magnetizations are of steeper inclination than J magnetizations, by about 6°. This could be interpreted to indicate modest (~3°±6.5°) latitudinal displacement between the Caborca terrane and the Jurassic arc, although the magnitude of displacement is smaller than the statistical uncertainty. Moreover, we emphasize that the null hypothesis of J and J* populations sharing a common mean cannot be rejected with 95% confidence using a modified *McFadden and Lowes* [1981] test for populations with different k values.

If the test of the MSM hypothesis was based purely on inclination differences north and south of the megashear, a 95% confidence interval of 6.5° would make our conclusions less definitive. However, declinations north and south of the MSM are identical, and we can reject the hypothesis of significant rotation between the Caborca terrane and the Jurassic Cordilleran arc with 95% confidence. Furthermore, the MSM model predicts a modest amount of counter-clockwise rotation and small (southward) latitudinal displacement of the Caborca block with respect to North America. Therefore pre-Late Jurassic magnetizations in the Caborca terrane should be characterized by inclinations steeper than those expected using the North America reference pole, and declinations for normal polarity magnetizations should be west of those expected. Instead, declinations deviate eastward from those expected for the corresponding interval that brackets the age of J and J* magnetizations, even after allowing for a small (less than about 10°) clockwise rotation in Late Cretaceous time. In turn, observed inclinations are concordant or shallower than expected.

For this analysis, J and J* site means from rocks north and south of the megashear were combined to obtain a mean of D=15.0°, I=8.5° (n=38 sites) and a pole at 61°N, 36°E (dp=3.0, dm=5.9). All sites from six localities are included in this average. Selected site means with n>2, k>10 and α_{95}<25, yield a mean of D=13.5°, I=9.0° (n=24 sites; k=18.6, α_{95}=71°) which is indistinguishable from the mean for the entire collection (Figure 16f). Expected declinations and inclinations for a central location in the Caborca region (30.5°N, 112°W), calculated for a set of high-quality paleopoles derived from rocks on the North American craton, are plotted against the age of the pole for the interval between 200 and 80 Ma and compared with our observed data. Error bars correspond to stratigraphic ranges or the uncertainty of isotopic age determinations, and the A_{95} of the reference pole. These error bars are thus overconservative. Most of the data were obtained from the global paleomagnetic data base [*McElhinny and Lock*, 1996] with additional data from more recent publications.

The configuration of the Jurassic segment of the North American APWP is controversial, not only because of the scarcity of results, but also because Middle Jurassic poles inferred to be of the same age produce contrasting results [*May and Butler*, 1986; *Van Fossen and Kent*, 1990]. The shaded area in Figure 17 encloses all the uncertainties in the paleomagnetic reference data. Our general conclusions about pre-

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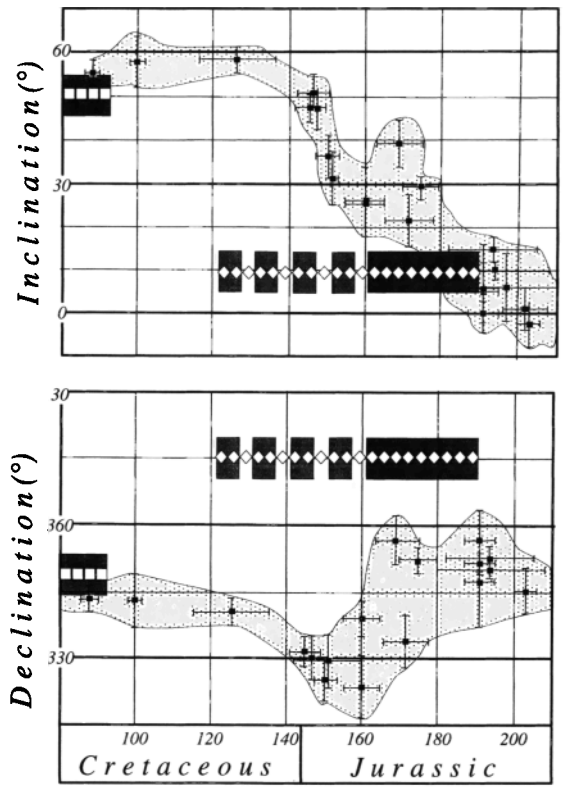


Figure 17. Observed Jurassic (diamonds) and Late Cretaceous (squares) declinations (lower panel) and inclinations (upper panel) for the Caborca region compared to the expected declinations and inclinations calculated from selected cratonic North America reference poles for the interval from about 200 to about 80 Ma. The box around the observed declination (or inclination) corresponds to the $\alpha 95$. The interpreted age range that brackets the Jurassic ChRM is 190 to 160 Ma, but an age as young as 120 Ma is also permissible. Error bars for the expected directions are based on the pole confidence region. Error bars also incorporate the stratigraphic ranges or errors associated with isotopic age determinations for the reference poles. The shaded area encloses the uncertainties in the predicted declinations and inclinations.

Early Cretaceous latitudinal displacement and rotation are independent of our choice of appropriate Middle Jurassic poles, although the actual rotation parameters estimated are obviously different. For the interval that brackets acquisition of J and J* magnetizations, rotation with respect to North America is exclusively clockwise, from 12° to as much as 50° (Figure 17). Estimates of latitudinal displacement vary from as little as 2.9°±6.3° southward, for a Sinemurian age (195 Ma) when the observed magnetizations are compared to the group of poles for the Kayenta Formation [*Bazard and Butler*, 1991], to as much as 7.9°±3.1 northward for a Callovian-Oxfordian age (155 Ma) when the observed magnetizations are compared with the pole for the Summerville Formation [*Bazard and Butler*, 1992]. Using high-latitude poles such as that of the Moat volcanic rocks [*Van Fossen and Kent*, 1990] produces even larger estimates of northward displacement. A significantly larger and statistically significant northward displacement would be calculated assuming a latest Jurassic-Early Cretaceous age for J* magnetizations. In a similar fashion, using only J* magnetizations for the Caborca block, and the oldest possible age for the time of remanence acquisition (post-Pliensbachian), the maximum displacement allowed by the data is 1.7°±4.0° (southward). This leads us to conclude that the hypothesis of southward displacement of more than 5.7° (~600 km) of the Caborca block with respect to the craton can be rejected with 95% confidence. In this calculation we use the synthetic pole position for 190 Ma after *Gordon et al.* [1984]. A higher-latitude pole such as the 190 Ma pole of *Irving and Irving* [1982] would indicate a minimum 3.2°±3.9° (~300 km) of northward latitudinal displacement of the Caborca terrane. Together, the lack of clear evidence for southward displacement, the observed clockwise rotation, and the similarity of the Jurassic magnetizations in the Cordilleran arc with those of the Caborca block are not consistent with the Mojave-Sonora megashear model of significant Late Jurassic southeast motion of northern Mexico along a left-lateral strike-slip fault zone.

Recently, other geologic data have weakened arguments previously used to support mid-Mesozoic displacement along the hypothetical MSM. The apparent abrupt termination of rocks of the Cordilleran Jurassic volcanic arc along a northwest trending line north of Caborca, interpreted to be the trace of the MSM (Figure 1b), may be an artifact of Cretaceous deformation [*Jacques-Ayala et al.*, 1990]. For example, *Calmus and Sosson* [1995] have shown that outcrops of Jurassic volcanic rocks extend about 75 km south of the inferred trace of the megashear to the Gulf of California. Jurassic arc-related volcanic rocks may also form part of Sierra Rajon and Sierra Santa Rosa, 40 km south of the inferred MSM trace [*Longoria and Perez*, 1979; *Hardy*, 1981; *Molina Garza et al.*, 1997]. Furthermore, although Triassic strata of the Antimonio terrane bear important biostratigraphic similarities with the Luning assemblage in western Nevada [*Stanley and Gonzalez-Leon*, 1995], the dissimilarities are far more important [*Lucas et al.*, 1997; *Gómez-Luna and Martínez*, 1997]. Also, detrital zircon populations in Cambrian strata of the Provedora Quartzite in the Caborca terrane and the Bolsa Quartzite, north of the inferred trace of the MSM near Cananea, are identical [*Kurtz et al.*, 1998].

The MSM hypothesis also has been questioned on the basis of observed continuity of Precambrian and Paleozoic cratonal and miocline strata across the Mojave Desert, along the inferred traced of the MSM [*Cameron*, 1981]. Stratigraphic trends of Triassic rocks over this region have been also interpreted to be continuous [*Walker and Wardlaw*, 1989]. Other proposals in support of the MSM hypothesis, such as the possible existence of (displaced?) sections of the Cordilleran Jurassic continental arc in north-central Mexico [*Jones et al.*, 1995], have not yet been sufficiently evaluated, but we note that marine strata at Cerro Basura indicate that a marine basin lay south of the Jurassic arc.

6. Conclusions

Neoproterozoic through Lower Jurassic strata of the Caborca region are characterized by two secondary magnetizations, one (K*) associated with Late Cretaceous magmatism (<~80Ma) and an older J* magnetization for which the timing of acquisition is bracketed between about 120 and 190 Ma.

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Combined, Late Cretaceous secondary magnetizations and primary magnetizations in Cretaceous intrusions yield a mean of D=348.1°, I=50.7° (N=10 localities, 47 selected sites; k=53.5, $\alpha 95$=6.7°). These data are interpreted to indicate relative stability of the Caborca terrane with respect to North America. Latitudinal displacement and rotation estimates of 3.4°±2.9° northward, and 6.5°±8.7° clockwise, indicate that relative motion between the Caborca terrane and North America, if any, is below the detection limit of paleomagnetism. The regional internal consistency of data for Late Cretaceous intrusions suggests that effects of Tertiary tilt or rotation about a vertical axis over the broad region sampled (~5000 km^2) are not substantial.

The secondary J* magnetizations yield a mean of D=15.0°, I=10.0° (α_{95}=5.8°; k=23.0; n=28 sites). Magnetizations in Neoproterozoic miogeoclinal strata and arc-derived strata of the Sierra Santa Rosa Formation share a common mean and are inferred to be of the same age. J* magnetizations fail a paleomagnetic fold test at two localities. The shallow, post-folding inclinations of these magnetizations are clearly inconsistent, however, with remagnetization after Late Cretaceous folding, the age of compressional deformation previously assumed for this region. Our interpretation of the timing of remanence acquisition of J* magnetizations implies that most compressional deformation in areas south and west of Caborca must be Jurassic or Early Cretaceous in age.

Rocks of the Jurassic Cordilleran volcanic arc give a mean of D=15.0°, I=4.0° (α_{95}=14.3°; k=12.4; n=10 sites). These magnetizations pass a fold test and are interpreted as primary in origin. The age of these rocks is roughly bracketed between 190 and 160 Ma. The mean directions for data from rocks south and north of the MSM are statistically indistinguishable, arguing against the existence of a major crustal discontinuity along the proposed location of the MSM. This conclusion, we note, is independent of the configuration of the Jurassic North American APWP.

For the interval that brackets acquisition of secondary J* magnetizations, rotation with respect to North America is 12° to 50° clockwise, depending on the age assumed for the remanence. Estimates of latitudinal displacement vary from as little as ~300 km southward, for a Sinemurian age (195 Ma), to as much as ~800 km northward for an Oxfordian age (155 Ma). High-latitude Middle Jurassic poles [*Van Fossen and Kent*, 1990] produce larger estimates of northward displacement (>1400 km). Although the timing of acquisition is based on reasonable geological arguments, an Early Cretaceous age for J* magnetizations is permissible. Such an interpretation would indicate significantly larger northward displacement (>2000 km) with respect to cratonic North America. Paleomagnetic data are therefore inconsistent with proposed southeastward displacement of the Caborca block along the MSM.

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